Trading Scope for Credibility in Difference-in-Differences

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Abstract

In staggered difference-in-differences, parallel trends may hold for some treated cohorts and fail for others, so the standard estimator is biased for the average treatment effect on the treated (ATT). Honest inference that keeps the ATT as the target does not repair this, reading a pooled pre-trend that the cohort-level violations can wash out of, so its confidence set is either wide or falsely narrow. We take a different route: drop the offending cohorts and target the effect for the credible ones, the difference-in-differences analogue of the local average treatment effect. This credible-subpopulation effect is point-identified under a weaker parallel-trends requirement that can hold when the ATT's fails. But reading the credible cohorts off the observed pre-trends revives the pre-test problem the honest-DiD literature was built to avoid, which existing methods do not cover because they never select. We formalize this change of estimand and supply honest inference for it, composing randomized selective inference (Fithian et al., 2014; Tian and Taylor, 2018) with the sensitivity bounds of Rambachan and Roth (2023) into an interval uniformly valid against both the data-driven selection and the residual violation of parallel trends the screen cannot rule out. We characterize in closed form when narrowing the target lowers risk. In an application to the shale boom, a relative-magnitudes analysis certifies a significantly positive pooled effect on house prices, but the credible subpopulation shows no effect, locating the pooled positive in cohorts already trending before onset.

Keywords. Difference-in-differences, Staggered adoption, Parallel trends, Robust inference, Sensitivity analysis, Partial identification.

JEL classification. C1, C23, C51.

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1 Introduction

Difference-in-differences is among the most widely used research designs in empirical economics, and its credibility rests on the parallel trends assumption, that absent treatment treated and comparison units would have evolved in parallel. Researchers assess it by testing for pre-treatment differences in trends, but that practice is fragile. Pre-trends tests are often underpowered against economically meaningful violations (Bilinski and Hatfield, 2020; Freyaldenhoven et al., 2019; Kahn-Lang and Lang, 2020; Roth, 2022). Conditioning on passing them induces a pre-test selection bias (Roth, 2022), though such conditional inference remains valid, if conservative, when parallel trends holds (de Chaisemartin and D'Haultfœuille, 2024). And even a clean pre-period does not guarantee a clean post-period, since nothing prevents a confound from arriving at the moment of treatment (Kahn-Lang and Lang, 2020).

A growing literature, surveyed in Roth et al. (2023), responds by relaxing parallel trends. Building on Manski and Pepper (2018), Rambachan and Roth (2023) bound how far post-treatment violations can depart from the observed pre-trends and report a uniformly valid confidence set for the ATT. Kwon and Roth (2024) sharpen the set with an empirical-Bayes prior on the violation, and de Chaisemartin (2026) rank the post-treatment differences against the pre-treatment ones. All keep the ATT and use the pre-trends to extrapolate the violation across periods. But the ATT is where a structural difficulty lies. When parallel trends fails for some cohorts and holds for others, the ATT is an average over all treated cohorts, including the offenders, and its point estimate is biased. Staggered adoption makes such heterogeneity the rule rather than the exception. Each cohort's timing must be defended on its own, one cohort's adoption perhaps as good as randomly timed while another's coincides with a confounding shock, so parallel trends is naturally a cohort-level property, credible for some and not for others. Honest inference that keeps the ATT as the target does not repair this. It reads a pre-trend that pools the cohorts, and the failure takes one of two forms. When the aggregate pre-trend reveals the violation, the honest set is valid but wide, accommodating the worst cohort. When the offending cohorts' pre-trends wash out of the aggregate, a minority, differently timed, the honest set is instead falsely precise, tight around the biased estimate. As Section 6 shows, a relative-magnitudes sensitivity analysis then certifies a spurious effect. Either way the fixed-target analysis on the aggregate is the wrong instrument, because the violation is a property of the cohort composition that the aggregate hides.

Our response is to change the target. When parallel trends fails for some cohorts, we drop them and report the effect for the credible ones. The obstacle is inference. Selecting the credible cohorts from the pre-trends revives the very pre-test bias honest-DiD exists to avoid, and no existing honest method addresses it, because none selects. This paper supplies that inference. Applied practice already retreats to the part of a design it trusts, but piecemeal and without inference to match. Researchers exclude treated units whose inclusion would contaminate the comparison, select and weight comparison units for pre-treatment fit (Abadie et al., 2010), and, less formally, adjust samples and specifications until the pre-trends look flat, the specification search whose statistical costs Roth (2022) documents. Each is a retreat to a more credible subpopulation, made ad hoc or on design grounds, and none carries inference valid against the selection it performs. When the population ATT is not credibly identified, we make that retreat explicit and give it a target and an inference. We drop to the subpopulation of cohorts where identification is credible, the logic of the local average treatment effect (Imbens and Angrist, 1994), of the optimal-subpopulation average treatment effect under limited overlap (Crump et al., 2009), and of overlap weighting (Li et al., 2018), each of which trades the original estimand for a credibly identified effect on a selected subpopulation. The credible-subpopulation LATT is a reweighting of standard group-time effects (Callaway and Sant'Anna, 2021) over the cohorts whose parallel trends is credible.[1] Credibility can be decided ex ante, from covariates or institutional knowledge, but in the leading case it is read from the pre-trends themselves, the same pre-trends used for estimation, which is exactly the selection the honest-DiD literature avoided. The paper thus makes two contributions. The first is to propose and formalize this change of estimand for staggered difference-in-differences, giving the credible-subpopulation effect an identification result and a rule for when to adopt it. The second, where the work lies, is to make the data-driven selection honest, with inference for the LATT that is uniformly valid against both the selection and the residual violation of parallel trends that a pre-trend screen cannot rule out.

The LATT is point-identified under parallel trends on the selected cohorts alone, a weaker requirement that can hold when the ATT's fails, and this identification is immediate. The work is inference. Because the screen selects on pre-trends, and a flat pre-trend does not certify the flat post-trend that identification requires, the honest interval, not the point, is the object we carry. This second contribution, the inference a data-driven selection demands, has three parts. First, and centrally, honest post-selection inference. We combine randomized selective inference (Tian and Taylor, 2018), in the data-carving framework of Fithian et al. (2014) and implemented through the polyhedral lemma of Lee et al. (2016), with the sensitivity bounds of Rambachan and Roth (2023) into a single interval for the LATT that is uniformly valid in the data-generating parameter, with no separation or consistent-selection condition when the event-study covariance is known, and to first order when it is consistently estimated. The carved component absorbs the data-driven selection and the level bound absorbs the residual identification failure, and the two compose. Neither ingredient suffices alone. Selective inference (Fithian et al., 2014; Lee et al., 2016; Tian and Taylor, 2018) delivers valid inference when the target is point-identified conditional on the selected set. Here it is not, so carving alone leaves a residual violation it cannot touch, and composing it with a sensitivity class is what restores honesty. This is the step the fixed-target sensitivity analysis never takes, because it never selects, and it carries selection and identification uncertainty on separate instruments. Second, a feasible decision rule. We characterize in closed form when the trade pays, when the credible-subpopulation honest interval is shorter than the honest interval for the ATT. A

[1] These estimators (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfœuille, 2020) repair the aggregation failure of two-way fixed effects under staggered adoption (Goodman-Bacon, 2021), whose implicit weights can turn negative under treatment-effect heterogeneity. That repair concerns aggregation, not identification, since they maintain parallel trends and are biased when it fails, which is the failure we address.

width-dominance theorem gives the explicit threshold on the dropped cohorts' violation above which the credible-subpopulation interval is strictly shorter than the honest ATT interval, a threshold estimable from the covariance and the chosen sensitivity bounds. The closed-form threshold is for the fixed-length intervals of ex-ante selection. Under data-driven selection the carved interval's expected length is infinite, so the comparison there is by coverage and median length rather than a width threshold. We turn the characterization into an estimable rule reported alongside both honest intervals. Because both intervals are honest, the rule can never cost coverage, only efficiency. Under homogeneous effects it selects the lower-risk procedure with probability approaching one, and under heterogeneity it returns a breakdown value Γ^* in the idiom of a sensitivity analysis. Third, evidence. Simulations trace the method's advantage across the informativeness of pre-trends and confirm the width dominance directly. Once the dropped violation is large enough, the honest credible-subpopulation interval is strictly shorter than the honest ATT interval, both covering. An application to the shale boom then withdraws a spuriously positive pooled estimate of the effect on house prices. A relative-magnitudes sensitivity analysis certifies a +6.7 log-point effect, yet the credible subpopulation, the counties whose onset did not coincide with the housing run-up, reveals essentially zero, locating the pooled effect in the cohorts already trending before onset. An appendix develops the optimal credible weighting that the same sensitivity class implies, a soft-thresholded weighting of cohorts by credibility net of noise, of which the flatness screen is a special case, completing the analogy to the variance-minimizing overlap weights of Crump et al. (2009). The most closely related prior work is de Chaisemartin (2026), which also brings the pre-trends into the inference rather than leaving them to a pre-test, but keeps the ATT and bounds it by ranking the post-treatment differences against the pre-treatment ones; we instead change the target to the credible subpopulation, so the object inference must be valid against is the data-driven cohort selection, not the extrapolation across periods.

The remainder of the paper is organized as follows. Section 2 fixes the staggered-DiD setup, defines the credible-subpopulation LATT, and develops its identification and estimation. Section 3 develops the honest inference that a data-driven selection demands, and Section 4 the decision of when to narrow the target. Section 5 reports the simulation study, Section 6 applies the method to the local house-price effect of the shale boom, and Section 7 concludes.

2 The credible-subpopulation effect

2.1 Setup

We adopt the staggered adoption framework of Callaway and Sant'Anna (2021). There are periods $t = 1, \ldots, T$ and units i indexed by their first treatment date $G_i \in \{2, \ldots, T\} \cup \{\infty\}$, where $G_i = \infty$ denotes never-treated. Let $Y_{it}(g)$ denote the potential outcome in period t if unit i is first treated in period g, and $Y_{it}(\infty)$ the never-treated potential outcome. The building-block causal parameter is the group-time average treatment effect on the treated,

$$\mathrm{ATT}(g,t) = \mathbb{E}[\,Y_{it}(g) - Y_{it}(\infty) \mid G_i = g\,]\,. \tag{1}$$

Summary parameters are weighted aggregations $\theta = \sum_g \sum_t w(g,t)\,\mathrm{ATT}(g,t)$ with researcher-chosen weights (Callaway and Sant'Anna, 2021), of which the overall ATT and the event-study path are special cases.

It is convenient to work with the event-study representation and the decomposition of Rambachan and Roth (2023). Let $\hat\beta = (\hat\beta_{\mathrm{pre}}', \hat\beta_{\mathrm{post}}')'$ be an asymptotically normal vector of event-study coefficients estimated by any of the heterogeneity-robust procedures above, with $\sqrt{n}(\hat\beta - \beta) \to \mathcal{N}(0, \Sigma)$. The finite-sample covariance of $\hat\beta$ is then $\Sigma_n = \Sigma/n$, and the Gaussian working model of Section 3, together with the standard errors, truncation limits, and randomization scale used throughout, is stated in terms of Σ_n, and we write Σ for Σ_n when no confusion arises. The estimand decomposes as

$$\beta = \tau + \delta, \qquad \tau_{\mathrm{pre}} = 0, \tag{2}$$

where τ collects the causal effects (zero before treatment, by no anticipation) and δ is the differential trend between treated and comparison groups that would have occurred absent treatment. Parallel trends is the restriction $\delta_{\mathrm{post}} = 0$, under which $\beta_{\mathrm{post}} = \tau_{\mathrm{post}}$, and a pre-trends test is a test of $\delta_{\mathrm{pre}} = 0$.

Assumption 1 (Maintained conditions)**.** *(i) Asymptotic normality. The event-study estimator satisfies $\sqrt{n}(\hat\beta - \beta) \xrightarrow{d} \mathcal{N}(0, \Sigma)$ for a positive-definite Σ, consistently estimable by some $\hat\Sigma \xrightarrow{p} \Sigma$. (ii) No anticipation. $\tau_{g,\mathrm{pre}} = 0$ for every cohort g, so that the pre-treatment coefficients identify the pre-treatment differential trend, $\beta_{g,\mathrm{pre}} = \delta_{g,\mathrm{pre}}$.*

All results maintain Assumption 1. The finite-sample carved inference of Section 3 additionally works in the Gaussian model $\hat\beta \sim \mathcal{N}(\beta, \Sigma)$ with Σ known, with the estimated-Σ case treated in Remark 4.

We will require the analogous objects at the cohort level. Write δ_g for the differential trend of cohort g and say that parallel trends holds for g if $\delta_{g,\mathrm{post}} = 0$. A key feature of applications, motivated in the introduction, is that this may hold for some cohorts and fail for others, since each cohort's adoption timing must be defended on its own.

Finally, we distinguish two regimes for how cohorts are judged credible, because they carry different inferential guarantees. Under ex-ante (or independent) selection, credibility is decided from pre-determined covariates, institutional knowledge, or an independent data split, so that the selected set is a fixed S statistically independent of the estimation errors in $\hat\beta$. Under data-driven selection, credibility is assessed from the realized pre-trends $\hat\beta_{\mathrm{pre}}$, so that the selected set $\hat S$ is random and depends on the estimation sample, with a deterministic population counterpart S^*, the set a population pre-trend statistic would select. The identification argument below holds for any set held fixed, and so identifies the random target $\theta_{\hat S}$ conditional on the realized $\hat S$. Identifying the fixed population parameter θ_{S^*} instead requires $\hat{S}$ to recover S^*, a selection-consistency condition we return to under inference (Remark 1). The distinction is thus minor for identification but central for inference, and we treat it explicitly below.

2.2 Estimand and identification

Let S denote a set of cohorts judged to have credible parallel trends, produced by a selection rule R that we specify below. We define the target as the treatment effect for that subpopulation.

Definition 1 (Credible-subpopulation LATT). *For cohort weights $w_g > 0$ (e.g. proportional to cohort size) and a within-cohort aggregation $a(\cdot)$, a linear functional of a cohort's post-treatment vector (a fixed event-time, selecting one coordinate, or an average over post-treatment periods), the credible-subpopulation local ATT, defined for any nonempty S, is*

$$\theta_S \;=\; \frac{\sum_{g\in S} w_g \,\mathrm{ATT}(g,\cdot)}{\sum_{g\in S} w_g}, \qquad \mathrm{ATT}(g,\cdot) \;=\; a(\tau_{g,\mathrm{post}})\,, \tag{3}$$

the aggregated causal effect of cohort g.

The target is identified under a weaker condition than the ATT, a benchmark we state before turning, in Section 3, to the honest interval that is the paper's actual object.

Proposition 1 (Identification benchmark). *If parallel trends holds for every $g \in S$, i.e. $\delta_{g,\mathrm{post}} = 0$ for all $g \in S$, then θ_S is point-identified by the reweighted aggregated group-time effects, $\theta_S = \big(\sum_{g\in S} w_g\, a(\beta_{g,\mathrm{post}})\big)/\sum_{g\in S} w_g$, and this holds irrespective of whether $\delta_{g,\mathrm{post}} \neq 0$ for cohorts $g \notin S$.*

Proof. See Appendix C. □

The identification in Proposition 1 is immediate. The substance is the honest inference for the data-driven selection this target invites, developed in Section 3. Changing the target to the credible subpopulation weakens the identifying requirement from parallel trends on all cohorts to parallel trends on S, which can hold when the former fails, at the cost of a narrower question whose subpopulation S must then be characterized. The weakening is genuine but conditional, and two things must be kept separate. The identifying content is the substantive restriction $\delta_{g,\mathrm{post}} = 0$ for $g \in S$, an assumption about post-treatment trends. The flatness screen of Section 2.3 is a statistical classification rule that supplies evidence for that restriction from the pre-period, not the restriction itself, since a cohort flat before treatment may still drift after it. Point identification of the causal θ_S thus holds only when the restriction does, and the screen cannot certify it. We therefore treat the point-identified θ_S as a benchmark and pair it with the honest sensitivity bounds developed below, which bound the residual post-treatment violation the screen leaves behind and, in doing so, return a set rather than a point when that violation is nonzero.

Narrowing the target carries an interpretability cost, and a standing reporting rule addresses it. The credible subpopulation is delimited by a statistical property, flat pre-trends, rather than by a substantive behavioral criterion, so it is less transparently interpretable than the compliers of the local average treatment effect (Imbens and Angrist, 1994), and under heterogeneous effects it answers a different question than the ATT for the full treated population. We handle this directly. The gap between the two is the composition gap Γ, unidentified because it depends on the effects of the discarded cohorts, so we always report both honest intervals with the breakdown value Γ^*, and the reader sees the full-population target alongside the credible one and how far the discarded cohorts' effects must differ to overturn the reading, and wherever the retained set admits a substantive description, in the application of Section 6, the counties whose oil-and-gas onset did not coincide with the mid-2000s housing run-up, lead with that description, which converts a cohort selected for a convenient statistical property into a subpopulation a reader can name, characterizing it by its covariates the way applied work characterizes the compliers of an instrumental-variables design (Abadie, 2003).

The construction uses staggered timing only to supply the groups. What it requires is a partition of the treated population into groups whose parallel trends can each be assessed. Adoption cohorts are the natural groups, and their credibility heterogeneity comes for free from the variation in when and why units adopt. In a non-staggered design the same construction applies with groups defined by covariates or geography, provided the groups differ in credibility and there are enough pre-periods to estimate each group's pre-trend rather than chase noise. The groups should accordingly be subpopulations large enough to assess, not individual units, whose pre-trends are too noisy to select on without reviving the pre-test bias in its most severe form.

2.3 Selection and estimation

The selection rule R maps the data to the set S, and its form fixes both which subpopulation θ_S describes and the inferential guarantees available. It takes two forms, matching the two regimes of Section 2.1. Under ex-ante selection, S is fixed before estimation from pre-determined covariates, institutional knowledge of which cohorts were plausibly confounded, or an independent data split. Under data-driven selection, credibility is read from the estimated pre-trends, and a cohort enters S when its estimated pre-trend is close to flat,

$$S \;=\; \big\{\, g : \max_{e<0} |\hat{\beta}_{g,\mathrm{pre}}(e)| \le c \,\big\}, \tag{4}$$

for a threshold c.

This is the direct reading of parallel trends. The assumption is that the treated-comparison difference would have stayed constant absent treatment, so a cohort is credible when its pre-treatment difference is close to constant, and the screen keeps exactly those. It is posed in the same currency as the honest inference of the next subsection, which bounds how far the post-treatment difference could have wandered from flat, so selection and inference test one object, the level of the parallel-trends violation.[2]

The threshold trades scope against credibility, with a lax c retaining more cohorts but admitting larger residual violations and a strict c the reverse. Because rule (4) reads S off the noisy estimates $\hat{\beta}_{\text{pre}}$, the selected set is random and dependent on the estimation sample, which is the source of the pre-test bias below and of the inference subtleties that follow it.

We do not recommend committing to a single c. The sensitivity interval of Section 2 is applied to whichever set c selects, and its bound M must dominate that set's residual post-treatment violation, so the effect of c on the required M runs through how far screening on flat pre-trends also flattens post-trends. When pre-trends are informative about post-treatment violations, a stricter c yields a cleaner set that a smaller M suffices to cover, while a laxer c buys scope at the cost of a larger M, so the threshold trades scope for precision rather than credibility for nothing. When they are not, tightening c need not shrink the post-treatment violation and the required M can be nonmonotone in c. Either way the honest object to report is the estimand together with its interval traced as a function of c, the selection analogue of the breakdown curve we report for M. A reader then sees the scope-credibility-precision frontier directly and need not trust a single cutoff or a presumed monotone mapping. When a single value is nonetheless wanted, we recommend calibrating it to negligibility rather than to significance, setting c to the largest pre-trend that would bias θ_S by less than a stated tolerance, in the spirit of the non-inferiority tests of Bilinski and Hatfield (2020), rather than to a critical value of a pre-trends test, which would reinherit the low power that motivates the paper. The application of Section 6 reports such a path.

The estimator $\hat{\theta}_S$ simply reweights, over the selected cohorts, the sample group-time effects of whichever heterogeneity-robust estimator one uses, and nothing below depends on that choice.

Proposition 2 (Consistency and efficiency)**.** *Under ex-ante selection and the regularity conditions of the underlying group-time estimator, $\hat{\theta}_S$ is consistent for θ_S. If in addition parallel trends holds for all cohorts, $\hat{\theta}_S$ is consistent for the ATT when either treatment effects are homogeneous or S comprises all cohorts, and otherwise for the selected-cohort LATT, which differs from the ATT by the gap between the selected- and full-population weighted averages of the cohort effects. In either case there is an efficiency loss relative to estimators that use all cohorts.*

The consistency clause of Proposition 2 invites a sharper question. When does narrowing the target cost nothing, in that the LATT and the ATT are the same object? Writing $\theta_g := \text{ATT}(g, \cdot)$, $\mathcal{G}$ for the full set of treated cohorts, and $\theta_{\mathcal{G}}$ for the ATT, the two coincide for a fixed selected set S, $\theta_S = \theta_{\mathcal{G}}$, exactly when the retained cohorts' weighted-average effect equals the full population's, $\sum_{g \in S} w_g \theta_g / \sum_{g \in S} w_g = \sum_{g \in \mathcal{G}} w_g \theta_g / \sum_{g \in \mathcal{G}} w_g$. In particular when no cohort is dropped, when

[2] A researcher willing to trust that a straight pre-trend would have continued linearly can relax flatness to approximate linearity, screening instead on the pre-trend's curvature (its second difference) and pairing it with the smoothness sensitivity class of Rambachan and Roth (2023) and a linear-extrapolation estimator (Armstrong and Kolesár, 2018). This buys the steep-but-straight cohorts that flatness discards, at the cost of the extrapolation assumption. The principle is the same in either currency. The screen should be posed in whichever functional the sensitivity class bounds, flatness with the level bound and curvature with the smoothness class, since level and curvature rank cohorts differently and select different sets.

effects are homogeneous, and more generally whenever selection is uninformative about effect size. The gap $\theta_S - \theta_G$ is the composition gap Γ that Proposition 8 isolates in Appendix B, the LATT's counterpart of the selection that separates a local from an average treatment effect (Imbens and Angrist, 1994; Li et al., 2018). Target equivalence and estimator convergence are distinct. Under data-driven selection the estimators can still converge, when no cohort is detectable and the whole population is retained, by Proposition 3, even where the targets would differ for any strict selection, while a strict but ignorable selection leaves the targets equal yet the finite-sample estimators differ by the efficiency cost of Proposition 2.

Under data-driven selection the rule (4) couples S to the estimation errors. A confounded cohort whose pre-trend noise happens to fall below c is admitted, and its differential trend enters $\hat{\theta}_S$. This is a pre-test selection bias in the sense of Roth (2022), with a definite source and a definite fate.

Proposition 3 (Pre-test bias)**.** *Under data-driven selection, $\hat{\theta}_S$ carries a finite-sample bias whose leading term is the weighted average of the admitted confounded cohorts' post-treatment differential trends, times their probability of admission. The exact bias also reflects the random weight normalization on the selected set and, when pre- and post-treatment estimation errors are correlated, a term from conditioning on the selection. Whether the leading term vanishes turns on detectability. Call a confounded cohort detectable if its population pre-trend statistic exceeds the threshold, $\max_{e<0}|\beta_{g,\text{pre}}(e)| > c$, and undetectable if it is flat in the pre-period, $\max_{e<0}|\beta_{g,\text{pre}}(e)| \leq c$, yet violates parallel trends in the post-period. As per-cohort information grows, a detectable cohort's admission probability tends to zero and its contribution to the bias vanishes, whereas an undetectable cohort's admission probability tends to one and its contribution persists at every sample size, while a cohort exactly at the threshold is admitted with an interior probability. Hence $\hat{\theta}_S$ is consistent for θ_S whenever the selected aggregate's residual violation vanishes, which holds in particular if every confounded cohort is detectable, though detectability of each is sufficient rather than necessary, since cohort-level violations may cancel in the aggregate. Otherwise $\hat{\theta}_S$ retains an asymptotic bias equal to the residual violation of the undetectable cohorts, the identification gap that the honest inference of the next subsection bounds.*

3 Honest inference after selection

A flat pre-trend does not guarantee a flat post-trend. Selection removes gross violations but not subtle residual ones. We therefore pair the point estimate with the sensitivity analysis of Rambachan and Roth (2023), applied to the selected-cohort aggregate. It imposes a restriction $\delta_{S,\text{post}} \in \Delta$ on the residual differential trend and reports the implied confidence set for θ_S. To match the flatness screen we use the level bound $\Delta^{\text{Level}}(M) = \{\delta : |\delta_{\text{post}}(e)| \leq M \ \forall e\}$, which asks how far the treated-comparison difference could have drifted from its flat pre-treatment level. The estimator is then the reweighted post-treatment average, with no extrapolation, and its fixed-length confidence interval (FLCI) is near-optimal (Armstrong and Kolesár, 2018). We use this imported construction throughout. For a scalar target estimated with standard error s under a worst-case bias bound B, the near-optimal FLCI of Armstrong and Kolesár (2018) has half-width

$$h(B,s) \;=\; s\,\mathrm{cv}_\alpha\big(B/s\big), \qquad \mathrm{cv}_\alpha(t) = \text{the } (1-\alpha) \text{ quantile of } |\mathcal{N}(t,1)|, \tag{5}$$

with $\mathrm{cv}_\alpha(0) = z_{1-\alpha/2}$ the pure-noise critical value and $\mathrm{cv}_\alpha(t) \uparrow t + z_{1-\alpha}$ as the bias comes to dominate. The bound M is not estimated but is the axis of a sensitivity analysis. The pre-period identifies δ_{pre} and not δ_{post}, so no screen can certify a value of M, and the honest object is not a single interval but the map from M to the interval it implies, together with the breakdown value M^*, the smallest post-treatment violation under which the stated conclusion no longer follows. This is the level-bound analogue of the breakdown reporting of Rambachan and Roth (2023). The honest object is not a claim that the retained cohorts' post-trends are flat, which no pre-trend can establish, but the map of how flat they must be for the reading to stand. Because M and M^* are in the outcome's own units, M^* is directly interpretable. We benchmark it two ways: against the retained cohorts' own pre-trends, which the screen holds below c in the same units, and against the effect at stake. An M^* several times the retained cohorts' visible pre-trends signals a conclusion surviving violations far larger than anything the pre-period displays. Section 6 reports both benchmarks.

Validity depends on the selection regime. Under ex-ante selection S is independent of $\hat\beta$, and the uniform validity of Rambachan and Roth (2023) applies verbatim. Under data-driven selection S is random, and two issues arise. The first is post-selection inference. It is asymptotically harmless when cohorts are well separated from the threshold.

Remark 1 (Pointwise oracle coverage under separation). A consistency argument already secures coverage under a strong condition. If the selection statistic is consistent and every cohort's population value lies a fixed distance from the threshold c (a separation condition), and the population credible set's residual violation satisfies $\delta_{S^*,\text{post}} \in \Delta^{\text{Level}}(M)$, then $P(\hat S = S^*) \to 1$ and the FLCI on the estimated set $\hat S$ inherits the oracle coverage of the interval on the population set S^*, nominal by the fixed-set validity noted above. This oracle guarantee has two limitations, both removed by the carved construction below. The separation condition is untestable and nearly restates the parallel-trends question, relocating the identifying content rather than supplying it, and the coverage is only pointwise rather than uniform, the local-to-threshold obstruction that Proposition 4 makes precise.

A researcher unwilling to assume separation has two options. One secures validity through independence, selecting on pre-determined covariates or an independent sample split, at the cost of the efficiency the split forgoes. The other conditions on the selection and recovers that efficiency. We develop it now, because it delivers the uniform guarantee the consistency argument lacks. The result rests on the following conditions and on the classical post-selection lemma, both collected here so the construction is self-contained.

Assumption 2 (Carved-inference conditions)**.** *In addition to Assumption 1, the finite-sample carved inference maintains. (i) the Gaussian model $\hat\beta \sim \mathcal{N}(\beta,\Sigma)$ with Σ known, the estimated-Σ case treated in Remark 4. (ii) the data-driven flatness screen R of* (4)*, randomized at a scale $\gamma \geq 0$, and* *(iii) conditioning on the realized selection event together with the active pre-period constraints and their signs, for each retained cohort the constraints that bind, for each dropped cohort a breaching coordinate and its sign.*

Lemma 1 (Polyhedral lemma, Lee et al., 2016)**.** *Let $y \sim \mathcal{N}(\mu, \Sigma)$ and fix a contrast η. On any polyhedral event $\{Ay \leq b\}$, decompose Ay into its component along $\eta' y$ and a remainder r that is independent of $\eta' y$. Conditional on the event $\{Ay \leq b\}$ and on the realized remainder r, the event confines $\eta' y$ to an interval $[\mathcal{V}^-, \mathcal{V}^+]$ whose endpoints are computable in closed form from (A, b, Σ, η) and r, and its law is $\mathcal{N}(\eta'\mu, \eta'\Sigma\eta)$ truncated to $[\mathcal{V}^-, \mathcal{V}^+]$. The truncated-normal pivot $F^{[\mathcal{V}^-,\mathcal{V}^+]}_{\eta'\mu,\,\eta'\Sigma\eta}(\eta' y)$ is therefore* Uniform$[0, 1]$ *conditional on the event and r, and, since this holds for every value of r, conditional on the event alone by iterated expectations.*

The construction randomizes the screen and conditions on its outcome, the randomized selective inference of Tian and Taylor (2018) in the data-carving framework of Fithian et al. (2014), with the exact truncation of each realized selection computed by the polyhedral lemma of Lee et al. (2016). The non-randomized limit $\gamma = 0$ is the exact polyhedral method of Lee et al. (2016); the randomization scale $\gamma > 0$ is the device of Tian and Taylor (2018), and an approximate-likelihood alternative is Panigrahi and Taylor (2023). In the reduced-form model $\hat{\beta} \sim \mathcal{N}(\beta, \Sigma)$ with Σ known, draw independent noise $\omega \sim \mathcal{N}(0, \gamma\, \Sigma_{\text{pre}})$ for a scale $\gamma \geq 0$, and select on the noised pre-trends, $\hat{S} = \{g : \max_e |\hat{\beta}_{g,\text{pre}}(e) + \omega_g(e)| \leq c\}$, while estimating the aggregate $\theta_S = \ell_S' \beta_{\text{post}}$ from the unrandomized $\hat{\theta}_S = \ell_S' \hat{\beta}_{\text{post}}$, where ℓ_S carries the reweighting onto S. The target is the reduced-form aggregate. Under parallel trends for S it is the causal θ_S, and otherwise it is that quantity plus the residual violation the level bound handles below. Selection uncertainty and identification are thus carried by separate instruments.

Theorem 1 (Uniformly valid carved inference)**.** *Maintain Assumption 2 and fix $\gamma \geq 0$. Conditioning on the selection event $\{\hat{S} = S\}$ together with the active flatness-screen coordinates and their signs renders the event a polyhedron in the Gaussian vector even with several pre-periods per cohort. Then, conditional on $\{\hat{S} = S\}$ and the Gaussian remainder r orthogonal to $\hat{\theta}_S$, the estimate has a truncated Gaussian law,*

$$\hat{\theta}_S \mid \{\hat{S} = S\}, r \ \sim\ \mathcal{TN}\big(\theta_S,\ s_S^2,\ [\mathcal{V}^-, \mathcal{V}^+]\big), \qquad s_S^2 = \ell_S' \Sigma_{\text{post}}\, \ell_S,$$

whose truncation limits $[\mathcal{V}^-, \mathcal{V}^+]$ are computable in closed form from Σ, γ, and the realized data by the polyhedral lemma of Lee et al. (2016). Inverting the truncated-normal pivot $F^{[\mathcal{V}^-,\mathcal{V}^+]}_{\theta_S, s_S^2}(\hat{\theta}_S)$ yields an interval $\mathcal{C}_{1-\alpha}(S)$ whose pivot is uniform for every realized r, hence uniform conditional on $\{\hat{S} = S\}$ by iterated expectations. This gives exact conditional coverage $\Pr(\theta_S \in \mathcal{C}_{1-\alpha}(S) \mid \hat{S} = S) = 1 - \alpha$ *for every $S \neq \varnothing$ and every β, and hence exact unconditional coverage* $\Pr(\theta_S \in \mathcal{C}_{1-\alpha}(\hat{S}) \mid \hat{S} \neq \varnothing) = 1 - \alpha$*, uniformly in β and without any separation condition.*

Proof. For a retained cohort the screen $\max_e |\hat{\beta}_{g,\text{pre}}(e) + \omega_g| \leq c$ is a box, an intersection of half-spaces. For a dropped cohort the complementary event $\max_e |\cdot| > c$ is a union of half-spaces and is not itself convex. Conditioning on a breaching coordinate e_g^* and its sign selects the single half-space $\pm(\hat{\beta}_{g,\text{pre}}(e_g^*) + \omega_g) > c$, and when several coordinates breach we condition on the full active set. The conditioned selection event is thus an intersection of half-spaces, a polyhedron in $(\hat{\beta}, \omega)$, for any number of pre-periods per cohort. This conditioning only refines the event, so the conditional pivot stays exact and unconditional coverage is recovered by averaging over the finer partition as well as over S. The pair $(\hat{\theta}_S$, selection) is jointly Gaussian, and decomposing the constraint vector into its component along $\hat{\theta}_S$ and an orthogonal remainder that is independent of $\hat{\theta}_S$, the event fixes the remainder and confines $\hat{\theta}_S$ to an interval $[\mathcal{V}^-, \mathcal{V}^+]$ whose endpoints are the tightest of the resulting linear bounds. This is Lemma 1 applied to the conditioned event. Hence the conditional law of $\hat{\theta}_S$ is $\mathcal{N}(\theta_S, s_S^2)$ truncated to that interval, the pivot is uniform on $[0, 1]$ conditional on $\hat{S} = S$, and inverting it gives an interval with conditional coverage $1 - \alpha$ for every S and every β. Averaging over S gives the same unconditional coverage, and since the finite-sample pivot is exact for known Σ regardless of where β sits relative to the threshold, coverage is uniform. The endpoint claims in γ hold because $\gamma = 0$ leaves the selection a deterministic function of $\hat{\beta}_{\text{pre}}$, while as $\gamma \to \infty$ the correlation between the selection variable and $\hat{\theta}_S$ vanishes, so $[\mathcal{V}^-, \mathcal{V}^+]$ expands to the real line and the truncated law returns to $\mathcal{N}(\theta_S, s_S^2)$. □

Remark 2 (The truncation limits and the role of γ). The limits $[\mathcal{V}^-, \mathcal{V}^+]$ are the intersection, over retained cohorts of the two-sided constraints $|\hat{\beta}_{g,\text{pre}} + \omega_g| \le c$ and over dropped cohorts of the one-sided active constraint $\pm(\hat{\beta}_{g,\text{pre}}(e_g^*) + \omega_g) > c$, of the intervals these place on $\hat{\theta}_S$ once r is held fixed. With a single pre-period per cohort, and when the dropped cohorts' screening statistics are uncorrelated with $\hat{\theta}_S$, the retained-cohort constraints alone bind; in general the dropped-cohort constraints also enter and are retained. The scale γ is a power parameter, not a validity parameter. At $\gamma = 0$ the interval is fully conditional; intermediate γ carves, recovering efficiency that full conditioning spends while retaining exact validity; and as $\gamma \to \infty$ the noise decorrelates the selection from $\hat{\theta}_S$ but sends every retention probability to zero, so $\hat{S} = \varnothing$ with probability approaching one. The useful regime is intermediate γ, and on the empty event the procedure reports the ATT interval.

Theorem 1 upgrades the pointwise oracle coverage of Remark 1 to a uniform guarantee, precisely across the local-to-threshold configurations that obstruct it. The guarantee is exact for known Σ and degrades to first-order validity when Σ is consistently estimated (Remark 4). That is the paper's single scope qualification, and we do not restate it elsewhere. It does so for the sampling and selection uncertainty in the reduced-form aggregate. The identification gap remains with the level bound, which we now fold in to reach the causal target.

Corollary 1 (Honest causal interval under data-driven selection)**.** *Let $\theta_{\hat{S}}^{\text{c}} = \ell_{\hat{S}}' \tau_{\text{post}}$ be the causal target on the selected set and $B_{\hat{S}} = \sup_{\delta_{\text{post}} \in \Delta^{\text{Level}}(M)} |\ell_{\hat{S}}' \delta_{\text{post}}|$ the worst-case residual bias the level bound allows, so that $|\theta_{\hat{S}} - \theta_{\hat{S}}^{\text{c}}| \le B_{\hat{S}}$ whenever $\delta_{\hat{S},\text{post}} \in \Delta^{\text{Level}}(M)$. For a single-period or convex-averaging functional $a(\cdot)$, $B_{\hat{S}} = M$. Then the widened carved interval*

$$\mathcal{C}_{1-\alpha}^{M}(\hat{S}) \;:=\; \left[\, \inf \mathcal{C}_{1-\alpha}(\hat{S}) - B_{\hat{S}},\; \sup \mathcal{C}_{1-\alpha}(\hat{S}) + B_{\hat{S}} \,\right]$$

covers the causal target, $\Pr\left(\theta^{c}_{\hat S} \in \mathcal{C}^{M}_{1-\alpha}(\hat S)\right) \geq 1-\alpha$, *uniformly in* β *and without a separation condition, whenever* $\delta_{\hat S,\text{post}} \in \Delta^{\text{Level}}(M)$.

Proof. See Appendix C. □

The widened interval $\mathcal{C}^{M}_{1-\alpha}(\hat S)$ is conservative. It superposes the exact carved interval for the reduced-form $\theta_{\hat S}$ on the worst-case additive bias $B_{\hat S}$ rather than folding sampling, selection, and identification uncertainty into a single length-optimal object, and is accordingly not the near-optimal fixed-length interval of Armstrong and Kolesár (2018). That construction and its optimality belong to the fixed-set case of Theorem 2, where selection does not randomize the target and the bias enters only through the standard error. Under data-driven selection the superposition buys a single interval valid against both the selection and the residual violation, at a modest excess length. The carved interval's exact coverage is the content of Theorem 1, and the multi-cohort panel study of Section 5.6 exhibits the understatement of naive uncertainty under estimated selection that carving is built to correct.

The second issue is that the bound M must accommodate a random selected set. The FLCI at M is valid only if the selected set's residual violation lies in $\Delta^{\text{Level}}(M)$. Since that violation is itself random, unconditional coverage requires M to dominate it with high probability.

Remark 3 (Calibration to the random selected set)*.* Under data-driven selection the selected aggregate's residual violation is random, so the level bound M may fail to dominate it. Write $\pi_M = \Pr\left(\delta_{\hat S,\text{post}} \notin \Delta^{\text{Level}}(M)\right)$ for the probability that the realized violation escapes the bound. Then unconditional coverage is at least $1-\alpha-\pi_M$. The carved interval covers the reduced-form $\theta_{\hat S}$ with probability $1-\alpha$ by Theorem 1, and its widening covers the causal target on the event that the bound holds, so $\{\theta_{\hat S} \in \mathcal{C}_{1-\alpha}(\hat S)\} \cap \{\delta_{\hat S,\text{post}} \in \Delta^{\text{Level}}(M)\} \subseteq \{\theta^{c}_{\hat S} \in \mathcal{C}^{M}_{1-\alpha}(\hat S)\}$ and a union bound gives the claim. Coverage thus approaches nominal only as $\pi_M \to 0$. Calibrating M to the mean residual violation typically leaves π_M bounded away from zero and undercovers. Taking M at an upper quantile of the residual-violation distribution makes π_M small but not zero, so it delivers coverage $1-\alpha-\pi_M$ rather than exactly $1-\alpha$ unless the error budget is widened to absorb π_M. That distribution is unidentified, which is why the reported object is the breakdown curve rather than a single M, read as a whole and with the breakdown value M^* read conservatively: a bound calibrated to the typical selected set would place a falsely narrow interval at the small-M end.

Finally, we caution against one otherwise-natural choice of Δ. The relative-magnitudes restriction $\Delta^{RM}(\bar M)$ of Rambachan and Roth (2023), following Manski and Pepper (2018), bounds post-treatment violations by $\bar M$ times the observed maximal pre-treatment violation, so its identified-set half-width is proportional to the observed pre-trend. When pre-trends are uninformative, they are small while the true violation need not be, so the identified set narrows toward a point even as the bias does not. Coverage then fails not from undercoverage within the maintained class but because the truth can lie outside it, the small observed pre-trend implying a relative-magnitudes class that excludes the true post-treatment violation. The fixed bound $\Delta^{\text{Level}}(M)$, set by the researcher rather than by the observed pre-trends, avoids tying the admissible violation to the pre-trend, though it too is honest only when its chosen M contains the true violation.

Remark 4 (Estimated covariance)*.* Theorem 1 treats Σ as known. In practice one plugs in a consistent $\hat{\Sigma}$, which enters twice, in the aggregate standard error s_S and in the selection geometry $[\mathcal{V}^-, \mathcal{V}^+]$ and carving scale, so it is more consequential here than in a fixed-target sensitivity analysis, where Σ enters only the standard error. If $\hat{\Sigma} \xrightarrow{p} \Sigma$ and the separation condition of Remark 1 holds, the truncation limits and scale are continuous in $\hat{\Sigma}$ and the selection event is correctly classified with probability approaching one, so by Slutsky the pivot converges to Uniform$[0, 1]$. The known-Σ exactness degrades to first-order valid coverage, which the panel simulation of Section 5.6, estimating the covariance from the data, bears out. This is the same feasibility standard the honest-DiD benchmark itself meets: the fixed-length interval of Rambachan and Roth (2023) likewise plugs in $\hat{\Sigma}$ and is uniformly valid only to first order. Under a separation condition the degradation is asymptotically inconsequential, and plug-in selective inference is uniformly asymptotically valid, so that the bootstrap may be used to calibrate it (Tibshirani et al., 2018). What separation buys, and what its absence costs, is worth stating precisely, because it marks the boundary of what any feasible procedure can deliver.

Fix a cohort whose population screening statistic sits local to the threshold, $\beta_g = c + h\,(s_g/\sqrt{n})$, so that the standardized distance h governs its retention probability, its truncation limits, and, through its inclusion, the identity of the estimand $\theta_{\hat{S}}$ itself. The whole difficulty of feasible inference is carried by this one scalar, which no estimator resolves.

Proposition 4 (Ceiling on feasible inference)**.** *Consider the local parametrization above with $\hat{\Sigma} \xrightarrow{p} \Sigma$.*

- *(i) (*Separation suffices.*) If $|h|$ is bounded away from zero uniformly, the selection event is classified correctly with probability approaching one, the plug-in truncation limits and scale converge, and the carved interval of Theorem 1 is uniformly asymptotically valid and first-order efficient (Tibshirani et al., 2018).*

- *(ii) (*Without separation, exact efficiency is unattainable.*) If h is left free, no feasible interval is simultaneously (a) uniformly asymptotically valid, (b) separation-free, and (c) asymptotically equal in expected length to the known-selection oracle. Near the threshold every uniformly valid feasible interval is strictly longer than the oracle; the excess length is irreducible.*

Sketch. The local parameter h is not consistently estimable: $\hat{h} = h + O_p(1)$, since $\hat{\beta}_g$ is $\sqrt{n}$-consistent for β_g and β_g approaches c at rate $1/\sqrt{n}$. The post-selection law is a discontinuous function of h at $h = 0$, so by Leeb and Pötscher (2005, 2008) it admits no uniformly consistent estimator; a feasible plug-in is therefore pointwise but not uniformly exact. Part (i) removes the discontinuity by keeping h bounded away from it, restoring uniform Slutsky continuity. Part (ii) is the oracle-length lower bound: recovering the known-selection length uniformly would require consistent classification of $\{\hat{S} = S\}$, hence consistent h, which is impossible; any uniformly valid feasible interval must instead hedge over the range of h consistent with the data, and this hedge has strictly positive length at $h = 0$. $\square$

The proposition delimits, rather than diminishes, what precedes it. The exact guarantee of Theorem 1 is a known-Σ benchmark; feasibly, separation recovers it asymptotically, and without separation one keeps uniform validity at a characterized cost in length. Only the single corner that is exact, efficient, uniform, and separation-free at once is closed off, and it is closed by a theorem, not by an unfinished derivation. A researcher unwilling to assume separation should therefore accept the near-threshold length inflation, hold the carving scale γ away from its degenerate limit, or use the independent split, whose validity rests on independence rather than on classifying the selection event and so survives estimated $\hat{\Sigma}$ without a separation condition, at a cost in power.

4 When the trade pays

The deliverable of this section is a rule the practitioner reports alongside both honest intervals. Prefer the credible-subpopulation LATT when narrowing lowers risk, and the ATT otherwise. Three forces set whether narrowing helps: the violation the screen removes, the variance that dropping a cohort costs, and the heterogeneity between the retained subpopulation and the full one. The sign of the advantage turns on their balance, not on the informativeness of the pre-trends, which governs only how much of a fixed-sign advantage is realized. The mean-squared-error account of these forces supplies the intuition and the explicit thresholds, the two-cohort law $V^2 > \gamma^2 + 2\sigma^2$, the estimable diagnostic $D^2 > \Delta\mathrm{Var}$, and the breakdown value Γ^* for the unidentified composition gap. It is developed in Appendix B, whose results we cite as needed. Here we give the operative test, the width comparison of the honest intervals themselves, since the intervals, not the point estimators, are the recommended output.

The comparison of honest intervals can be made exact in a two-cohort model, in the informative regime where the screen discards the confounded cohort with probability approaching one, so that the LATT selects the clean cohort alone. This is the regime in which the point-estimate advantage attains its ceiling (Appendix B), and it is where the method is meant to help. Both procedures report the level-bound fixed-length interval (5). The LATT, using the clean cohort alone, is unbiased with standard error σ and a residual-violation bound M_L, giving half-width $\sigma\,\mathrm{cv}_\alpha(M_L/\sigma)$. The ATT estimator, using both cohorts, has standard error $\sigma/\sqrt{2}$ but must set its bound to cover its own aggregate violation, $M_A = V/2$, giving half-width $(\sigma/\sqrt{2})\,\mathrm{cv}_\alpha(V/(\sqrt{2}\,\sigma))$. Under homogeneous effects, so that the clean-cohort LATT and the full ATT share a common target, each interval covers that target at its nominal level by construction, since each bound dominates its own aggregate's violation, so the honest comparison is not one of coverage but of expected length at equal coverage. Under heterogeneity the two intervals honestly cover different objects, and the comparison is one of length only once the composition gap is folded into the LATT interval evaluated as inference for the ATT.

We state the comparison in general, since it holds at any cohort count and isolates the single quantity, the violation carried by the dropped cohorts, that decides it. Write the ATT estimator as $\hat{\theta}_{\mathcal{G}} = \ell'_{\mathcal{G}}\hat{\beta}_{\mathrm{post}}$ with aggregate standard error $s^2_{\mathcal{G}} = \ell'_{\mathcal{G}}\Sigma_{\mathrm{post}}\ell_{\mathcal{G}}$ and honest bias bound $B_{\mathcal{G}}$ dominating the full set's violation $|\ell'_{\mathcal{G}}\delta_{\text{post}}|$, and the LATT estimator as $\hat{\theta}_S = \ell'_S\hat{\beta}_{\text{post}}$ with carved standard error s_S and residual bound B_S. Both honest intervals are the fixed-length construction (5).

Theorem 2 (Width dominance)**.** *Suppose both honest intervals are the fixed-length construction* (5)*, the LATT interval under ex-ante selection, equivalently the Rambachan and Roth (2023) interval on the retained set, with half-width $h_S = h(B_S, s_S)$, and the ATT interval $h_{\mathcal{G}} = h(B_{\mathcal{G}}, s_{\mathcal{G}})$. Then*

$$h_S < h_{\mathcal{G}} \quad \Longleftrightarrow \quad s_S\,\mathrm{cv}_\alpha\big(B_S/s_S\big) \;<\; s_{\mathcal{G}}\,\mathrm{cv}_\alpha\big(B_{\mathcal{G}}/s_{\mathcal{G}}\big).$$

Holding $(s_{\mathcal{G}}, s_S, B_S)$ fixed, if $s_S \geq s_{\mathcal{G}}$ there is a unique threshold

$$B^*_{\mathcal{G}} \;=\; s_{\mathcal{G}}\,\mathrm{cv}_\alpha^{-1}\Big(\tfrac{s_S}{s_{\mathcal{G}}}\,\mathrm{cv}_\alpha\big(B_S/s_S\big)\Big)$$

such that the credible-subpopulation interval is strictly shorter than the ATT interval if and only if $B_{\mathcal{G}} > B^*_{\mathcal{G}}$. If instead $s_S < s_{\mathcal{G}}$, the LATT interval can be shorter already at $B_{\mathcal{G}} = 0$ and no positive crossing threshold exists. Since $B_{\mathcal{G}}$ carries the dropped cohorts' aggregate violation $B_{\mathcal{F}}$ while B_S does not, this is a threshold on $B_{\mathcal{F}}$, and when restricting to S raises the aggregate variance, $s_S > s_{\mathcal{G}}$, the threshold is strictly positive, so a minimum dropped violation is required before bias relief outweighs the variance cost.

Proof. See Appendix C. □

Under data-driven selection the LATT interval is instead the carved interval of Theorem 1, whose truncated-normal inversion gives a data-dependent, generally asymmetric length rather than the fixed half-width $h(B_S, s_S)$. Here the width comparison does not extend, and for a definite reason. The carved interval inverts a truncated-normal pivot, and by Kivaranovic and Leeb (2021) such an interval has *infinite* conditional expected length, so an expected-length threshold is not merely unproven but ill-posed; Proposition 5 records this. The honest data-driven comparison is therefore by coverage, both intervals valid by Theorem 1 and Corollary 1, together with a robust length summary such as the median, which the simulation of Section 5.6 reports. An approximate-likelihood construction (Panigrahi and Taylor, 2023) would trade the exactness of the truncated-normal pivot for finite expected length; we retain the exact interval and summarize its length by the median. The fixed-length statement of Theorem 2 is accordingly stated for ex-ante selection, where the intervals are the fixed-length construction (5) and the threshold is exact.

Proposition 5 (No expected-length dominance under data-driven selection)**.** *Maintain Assumption 2. Under data-driven selection the carved interval $\mathcal{C}_{1-\alpha}(\hat{S})$ of Theorem 1 is the polyhedral inversion of the truncated-normal pivot $F^{[\mathcal{V}^-,\mathcal{V}^+]}_{\theta_S, s_S^2}$, and for every β and every $S \neq \varnothing$,*

$$\mathbb{E}\big[\,|\mathcal{C}_{1-\alpha}(S)|\;\big|\;\hat{S} = S\,\big] \;=\; \infty.$$

Consequently no expected-length ordering between the carved LATT interval and the ATT interval exists, and the width comparison of Theorem 2 is well-posed only for the ex-ante fixed-length intervals.

The randomization scale does not remove this, since every finite γ conditions on $\{\hat{S} = S\}$ and the active screen coordinates, leaving the pivot truncated normal.

Proof. See Appendix C. $\square$

Specializing to equal weights makes the threshold explicit. Let K equally weighted cohorts with common per-cohort post-treatment variance σ^2 and independent estimates comprise k clean cohorts ($V_g = 0$), retained, and $K - k$ confounded cohorts (violation V), dropped in the informative limit, with the retained set truly flat, $B_S = 0$. Writing $f = (K - k)/K$, so that $s_{\mathcal{G}} = \sigma/\sqrt{K}$, $s_S = \sigma/\sqrt{k}$, and $B_{\mathcal{G}} = fV$,

$$h_S < h_{\mathcal{G}} \quad \Longleftrightarrow \quad \mathrm{cv}_\alpha\Big(\frac{fV\sqrt{K}}{\sigma}\Big) > \sqrt{\tfrac{K}{k}}\, z_{1-\alpha/2} \quad \Longleftrightarrow \quad V > V^* := \frac{\sigma}{f\sqrt{K}}\, \mathrm{cv}_\alpha^{-1}\Big(\sqrt{\tfrac{K}{k}}\, z_{1-\alpha/2}\Big),$$

equivalently $B_{\mathcal{F}} = fV > B^*_{\mathcal{F}} = (\sigma/\sqrt{K})\, \mathrm{cv}_\alpha^{-1}\big(\sqrt{K/k}\, z_{1-\alpha/2}\big)$. At $K = 2$, $k = 1$ this is $V > 1.59\,\sigma$ (at $\alpha = 0.05$).

Two features carry the economics. The width threshold $V > 1.59\,\sigma$ exceeds the mean-squared-error threshold $V > \sqrt{2}\,\sigma \approx 1.41\,\sigma$ of Remark 5, because the fixed-length interval charges for the standard-error inflation of dropping a cohort, not only for its variance, and coverage is never the margin, since both procedures are honest, the ATT pays for honesty in a width that grows linearly in V, whereas the LATT pays a fixed $2\sigma\, z_{1-\alpha/2}$ set by its variance alone, the price-of-honesty reading of Figure 2. Allowing the retained set a residual bound $M_L > 0$ rather than exact flatness, the LATT interval carries standard error σ and bound M_L against the ATT's $\sigma/\sqrt{2}$ and $M_A = V/2$, so by (5) it is shorter if and only if $\mathrm{cv}_\alpha(M_L/\sigma) < \frac{1}{\sqrt{2}}\, \mathrm{cv}_\alpha(V/(\sqrt{2}\,\sigma))$. A residual $M_L > 0$, and heterogeneity $\gamma \neq 0$ when the target is the ATT (which adds $|\gamma|/2$ to the LATT's effective bias), both raise the threshold, in the direction the point analysis already indicated.

The width comparison assembles into a rule the practitioner can run. Report both honest intervals, for the ATT and for the LATT, and prefer whichever is shorter. Because each interval covers its own target whatever the data say, ranking them can misfire on efficiency but never on coverage.

Proposition 6 (A feasible and honest switching rule)**.** *Suppose the researcher reports the honest level-bound interval for both the ATT and the LATT, with half-widths $h_{\mathcal{G}} = h(B_{\mathcal{G}}, s_{\mathcal{G}})$ and $h_S = h(B_S, s_S)$ of (5), together with the rule. Prefer the LATT when its honest interval is strictly shorter, $h_S < h_{\mathcal{G}}$, and the ATT otherwise. Then the following hold.*

- *(i) (Honesty is unconditional.) Each reported interval covers its own target at level $1 - \alpha$ for every β, by the validity of the level-bound interval (carved, under data-driven selection, by Theorem 1). Since both are always reported, the rule, however it decides, cannot affect coverage. It ranks the two honest intervals and is an efficiency device only.*
- *(ii) (Feasibility.) The rule is computable from $\hat{\Sigma}$ and the chosen bounds alone. By Theorem 2, $h_S < h_{\mathcal{G}}$ if and only if the ATT's honest bound $B_{\mathcal{G}}$, which carries the dropped cohorts' aggregate*

violation $B_{\mathcal{F}}$, exceeds the threshold $B^*_{\mathcal{G}}$, so a minimum dropped violation is required before the LATT interval is the shorter one. The gap between the two point estimators $\hat{D} = \hat{\theta}_{\mathcal{G}} - \hat{\theta}_{\hat{S}}$, testable with the carved standard error of Theorem 1 applied to the contrast $(\ell_{\mathcal{G}} - \ell_S)'\hat{\beta}_{\text{post}}$, consistently estimates that dropped violation net of the composition gap in the informative limit, when the retained set's own residual violation is negligible.

(iii) (Sensitivity under heterogeneity.) The two intervals target different objects when the retained and discarded cohorts differ in effect, so the ranking can misfire on efficiency but never on coverage. The one quantity the data cannot identify is the composition gap $\Gamma = \theta_{\hat{S}} - \theta_{\mathcal{G}}$. We report it as a breakdown Γ^* in the same idiom as M^*, the amount by which the discarded cohorts' effects must differ from the retained cohorts' to overturn the reading, with its explicit form $\Gamma^* = (\Delta\text{Var} - D^2)/(2D)$ derived in Appendix B. This is a signed threshold defined for $D \neq 0$, with dominance requiring $\Gamma > \Gamma^*$ when $D > 0$ and $\Gamma < \Gamma^*$ when $D < 0$. At $D = 0$ the advantage is $-\Delta\text{Var} \leq 0$, so no finite breakdown exists, and the plug-in $\hat{\Gamma}^*$ is unstable when $\hat{D}$ is near zero.

Proof. See Appendix C. □

Proposition 6 turns the informal "report the LATT when pre-trends are informative" into a rule the data can run, and separates what they decide from what they cannot. The width comparison and $\hat{D}$ are estimable, while the unidentified composition gap is reported as the breakdown Γ^*, read as M^* is for the identifying restriction. The rule never suppresses an interval, so it cannot cost coverage.

Assembled in one place, the procedure is given in Algorithm 1.

Algorithm 1 The credible-subpopulation procedure

1. Estimate cohort-level (group-time) event studies with any heterogeneity-robust estimator (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfœuille, 2020).
2. Select the credible cohorts, from institutional knowledge or a covariate split (ex ante), or by screening the pre-trends for flatness (data-driven).
3. Estimate the LATT by reweighting the retained cohorts' post-treatment effects.
4. Report honest inference, the level-bound sensitivity interval on the retained set, carved for the selection when it is data-driven (Theorem 1), together with the breakdown violation M^*.
5. Report the trade, both honest intervals (LATT and ATT), preferring whichever is shorter by the width comparison (Theorem 2), with the breakdown value Γ^* for the unidentified composition gap (Proposition 6).

5 Simulation study

5.1 Design

We study the method in two environments. In the reduced-form model we draw event-study coefficients directly from their asymptotic distribution, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with Σ known, which isolates the identification and inference mechanics in the setting where the underlying theory is exact. In the panel model we generate individual outcomes for a staggered design and estimate both the coefficients and their covariance from the data, which confirms that the conclusions survive realistic estimation. Unless stated otherwise, results are averages over eight to twenty thousand replications.

The cohort structure follows a common template. Some cohorts are clean, with $\delta_g = 0$, and the remainder confounded, with a nonzero differential trend, and Table 1 gives the count and clean-confounded split used in each exercise. Every cohort is given the same true treatment effect, normalized to one. This normalization is deliberate, making the true ATT and the true LATT both equal to one, so that any departure of an estimator from one is attributable to a violation of parallel trends rather than to treatment-effect heterogeneity. It therefore isolates the identification problem that the paper is about.

Throughout, each cohort carries a four-period pre-treatment and four-period post-treatment event study with the reference period normalized to zero. A confounded cohort carries a level violation, its treated-comparison difference shifting by $V > 0$ in the post-period and by ϕV in the pre-period, for a scalar $\phi \geq 0$ we call the informativeness parameter and that is the master axis of the study. When $\phi = 0$ the confound is flat before treatment and no pre-trend rule can detect it, and as ϕ grows the pre-period shift announces it and it becomes detectable. Because V is the post-treatment violation it is directly comparable to the level bound M of $\Delta^{\text{Level}}(M)$. In the controlled coverage exercise below the aggregate being tested carries this violation exactly, which makes the boundary statement "covers if and only if $M \geq V$" checkable. More generally, coverage requires only that M dominate the selected aggregate's violation, which for a set mixing clean and confounded cohorts is weakly below the largest single-cohort V, so $M \geq V$ is sufficient and is necessary only in the worst case of an all-confounded set. This also makes ϕ the operational version of the informativeness of pre-trends discussed in Section 2.

The selection rule is the flatness screen, posed in the same currency as the honest inference, so cohort g is retained if its estimated pre-trend is close to flat, $\max_e |\hat{\beta}_{g,\text{pre}}(e)| \leq c$. The estimator of the ATT averages the post-treatment effect over all cohorts, as a heterogeneity-robust estimator would, and the LATT averages it over the selected cohorts only. Because the screen reads the selected set off the same noisy pre-trends, the selection is genuinely data-dependent in the sense of Section 2, and the pre-test bias of Proposition 3 is operative.

Table 1: Simulation parameters. Every exercise uses a four-period pre and four-period post event study with reference period zero, equal cohort weights, and true effect one for every cohort. The reduced-form model draws coefficients directly, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with within-cohort AR(1) covariance $\Sigma_{ij} = s^2\rho^{|i-j|}$. The panel model draws individual untreated outcomes $Y_{it}(0) = \delta_g(e) + \epsilon_{it}$, $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$, for 500 treated units per cohort and 1500 shared controls over $T = 14$ periods, and estimates the coefficients and their covariance from the data.

	Estimand (Table 2)	Scope (Fig. 1)	Honest cov. (Table 3, Fig. 2)	Panel (Sec. 5.6)
Cohorts (clean/confounded)	6 (4/2)	12 (6/6)	,	6 (3/3)
Post-treatment violation V	0.8	0.6	0.3, 0.6	0.6
Informativeness ϕ	1.0	$[0, 1.5]$	,	$[0, 1.5]$
Screen threshold c	0.40	0.40	,	0.40
Noise	s=0.03–0.40	σ=0.20	σ_{agg}=0.10	σ_ϵ=0.5
Pre/post correlation ρ	0.5	0.5	,	estimated
Replications	20,000	8,000	4,000	2000–3000

5.2 Recovering a clean effect where the ATT is biased

We first illustrate Proposition 1 in the simplest configuration, with a subset of cohorts violating parallel trends in the post-period. Table 2 reports the result. The estimator of the ATT is biased by +0.267, since it averages over all cohorts, so the offending cohorts' differential trends enter it directly. The LATT lands at 1.000, on the truth, because the selection removes those cohorts from the average and, by Proposition 1, their violations then do not enter the estimand at all.

The lower panel of the table sweeps the sampling noise s. As precision improves the LATT's bias falls monotonically from +0.013 to zero, and this residual is the selection-induced bias of Proposition 3, which vanishes as the selection rule becomes reliable. Two sources could in principle generate it, a noisy screen that admits genuinely confounded cohorts, which would persist even under independent selection, and the correlation between a cohort's pre- and post-treatment sampling errors, which distorts the retained cohorts' estimates only under same-sample selection. The final column isolates the two by selecting on an independent pre-trend draw. It tracks the full-sample column almost exactly, so the residual is the first source, admission of confounded cohorts, and the second is negligible here because the two-sided flatness screen conditions the retained estimates symmetrically. The persistent component is thus a residual identification bias, present even under independent selection and addressed by the sensitivity widening of the next section, rather than the strict post-selection bias that carving removes. The ATT estimator's bias, by contrast, is frozen at +0.267 regardless of precision. The LATT faces a noise problem, which more data cures, whereas the ATT estimator faces a wrong-target problem, which no amount of data cures.

5.3 The scope condition

Figure 1 traces the method across the informativeness axis ϕ described in Section 5.1.

The upper-left panel plots the bias of the two point estimators against the truth. The bias of the

Table 2: The estimand demonstration (reduced-form model). True effect = 1 for all cohorts. In the lower panel LATT (full) selects on the same pre-trends used for estimation, while LATT (split) selects on an independent draw. Their near-equality shows the residual bias is admission of confounded cohorts, not same-sample conditioning.

	ATT	LATT (full)	LATT (split)
Point estimate	1.267	1.000	,
Bias vs. truth	+0.267	+0.000	,
Residual LATT bias as sampling noise s falls			
$s = 0.40$	+0.265	+0.013	+0.014
$s = 0.20$	+0.267	−0.001	−0.000
$s = 0.12$	+0.267	−0.000	−0.000
$s = 0.03$	+0.267	+0.000	+0.000

ATT estimator is a flat line at +0.30, since it never consults pre-trends, so its performance cannot depend on how informative they are. The LATT's bias begins on top of that line at $\phi = 0$ and falls to zero as pre-trends become informative. The vertical gap between the two curves is therefore exactly the method's advantage, and reading it off from left to right gives the scope condition, with the advantage nil when pre-trends carry no information about post-treatment violations and maximal when they carry a great deal.

The two curves coincide at the left edge because selecting on uninformative pre-trends buys nothing, as Section 2 argues.

The upper-right panel explains why the LATT's bias falls, by plotting the identification gap of the selected set, the average post-treatment violation among the cohorts that survive selection. It tracks the LATT's bias almost exactly, falling from 0.30 to essentially zero. This confirms that the LATT's remaining bias is not an artifact of the estimator but the residual violation that selection has failed to remove. The lower-right panel gives the mechanism behind both, the average number of confounded cohorts that slip through the screen falling from 5.09 of six when the confound is invisible to essentially zero when it is plainly visible.

The lower-left panel turns to inference and reports the coverage of a point-based confidence interval for the causal target, computed both in the naive way and by sample-splitting. Both collapse at low informativeness and recover only at high informativeness. The comparison is diagnostic, since sample-splitting removes any contamination from data-dependent selection by construction, so the fact that it collapses just as badly establishes that the failure is not selection distortion but the identification gap of the upper-right panel, since the surviving cohorts genuinely still violate parallel trends. This is what motivates the sensitivity bounds of the next subsection, since no purely inferential fix can repair a violation of the identifying assumption.

(At the right edge the naive interval slightly outperforms the split one. This is because splitting discards half the data, and by that point selection is nearly deterministic, so there is no contamination left to correct.)

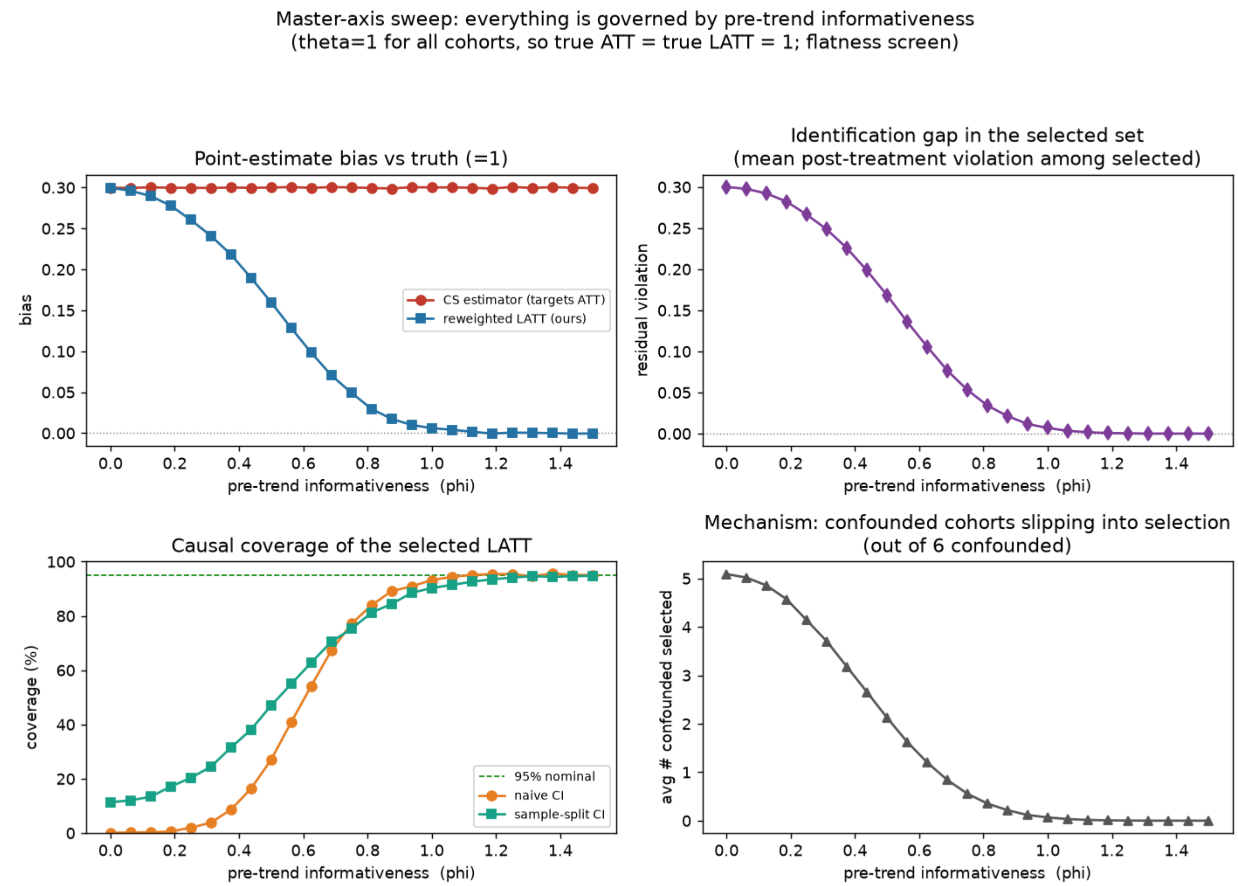

Figure 1: Scope condition. Everything is governed by pre-trend informativeness. The ATT estimator's bias is constant, and the LATT's advantage is the gap between the lines. Causal-coverage collapse at low informativeness is an identification gap (surviving cohorts still violate parallel trends), not selection distortion, since sample-splitting collapses just as badly.

5.4 Honest inference restores coverage

We now validate the level-bound FLCI of Section 2 applied to the selected-cohort aggregate, using the level violation of Section 5.1 whose size V is directly comparable to the assumed bound M.

Table 3 is the validity check. The FLCI is honest exactly on the range it claims, nominal at $V = M$, conservative when $M > V$, collapsing when $M < V$, failing visibly outside its bound rather than silently. The naive point interval, by contrast, collapses to near-zero coverage the instant any violation is present, since it is centered on a biased estimate with a width reflecting sampling noise alone.

Figure 2 repeats the exercise across a continuum of violation sizes and adds the practical output. In the left panel the point interval falls away rapidly as the violation grows, while each FLCI holds its nominal level until the true violation reaches its own assumed bound, marked by the vertical dotted lines, and declines thereafter. The middle panel records the price. Interval half-width is governed by the assumed M rather than by the realized violation, so honesty about a wider class of

Table 3: FLCI coverage of the causal target (reduced-form model), for an aggregate whose violation equals V, valid iff $M \geq V$.

True violation V	Assumed M	FLCI coverage	Point-estimate coverage
0.30	0.30 $(= V)$	95.0%	15.9%
0.30	0.60 $(> V)$	100%	14.9%
0.60	0.30 $(< V)$	9.3%	0.0%
0.60	0.60 $(= V)$	95.0%	0.0%

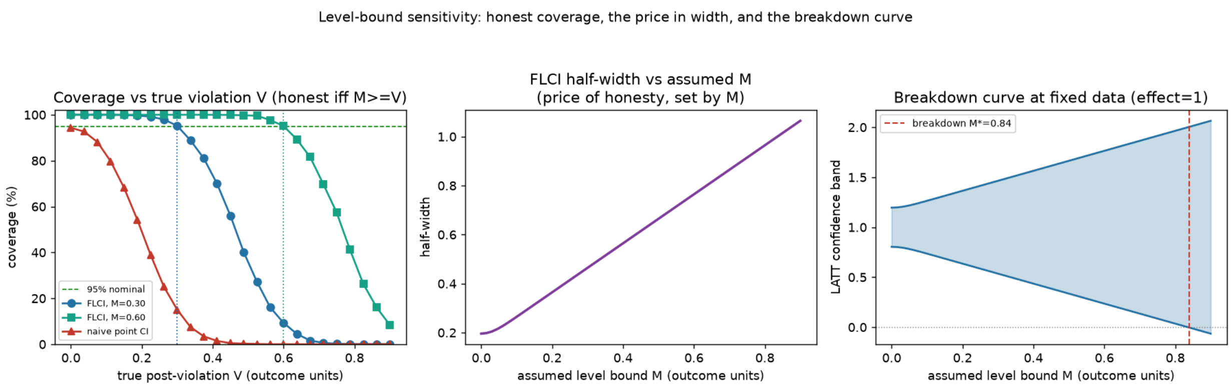

Figure 2: Sensitivity bounds restore honest coverage. The point estimate collapses as the violation grows. The FLCI is honest exactly while $M \geq V$. Interval width is the price of honesty, set by M. The right panel is the sensitivity/breakdown curve, with M^* in outcome units.

violations is paid for in width, immediately and proportionately.

The right panel is what a practitioner would actually report. Holding the data fixed, it traces the confidence band as the assumed bound is relaxed and identifies the breakdown value M^*. In this simulation, where the truth is known, M^* is the bound at which the band first admits the true effect. In an application, where it is not, the operative breakdown is the smallest M at which the band no longer supports the stated conclusion, for instance first includes zero, and the two need not coincide. Because M is in the outcome's own units, this quantity answers the question a reader of a DiD paper wants answered directly, how large a post-treatment parallel-trends violation one must be willing to countenance before the stated conclusion no longer follows. And because the bound is fixed by the researcher rather than read off the pre-trends, the exercise remains informative even when pre-trends are uninformative, in direct contrast to the relative-magnitudes restriction, whose identified set would shrink toward zero in exactly that case, leaving the truth able to fall outside the maintained class.

5.5 Width dominance and the shorter honest interval

Having confirmed that the honest interval covers, we return to the decision of Section 4. when is it actually shorter than the ATT's? Theorem 2 answered this analytically, with the explicit equal-weights threshold $V > 1.59\,\sigma$ at $K = 2$, $k = 1$. We confirm it directly and check the point that

Table 4: Width dominance ($K = 2$, $k = 1$, $\sigma = 1$, 40,000 replications). Both honest intervals cover the target at the nominal 95%. Past $V^* = 1.59$ the LATT half-width $h_{\mathcal{S}}$ undercuts the ATT's $h_{\mathcal{G}}$ at equal coverage, while the naive ATT interval is narrower only by undercovering.

V/σ	$h_{\mathcal{S}}$	$h_{\mathcal{G}}$	LATT cov.	ATT cov.	naive cov.
0.80	1.96	1.58	94.9%	94.9%	91.2%
1.59	1.96	1.96	94.9%	95.0%	79.6%
2.00	1.96	2.16	94.9%	95.0%	70.9%
2.50	1.96	2.41	94.9%	95.0%	57.8%

makes the comparison meaningful. The width gain is between two valid intervals, not an artifact of undercoverage.

The design is the informative limit of Section 4. Each of K equally weighted cohorts contributes an independent post-treatment aggregate $\hat{\beta}_g \sim \mathcal{N}(\tau_g + V_g, \sigma^2)$, with $\tau_g = 1$ so that the ATT and the LATT are both one and any width difference is attributable to the violation alone. The k clean cohorts ($V_g = 0$) are retained and the $K - k$ confounded ones ($V_g = V$) are dropped, ex ante, so both intervals are ordinary level-bound FLCIs (Armstrong and Kolesár, 2018). The LATT's, on the flat retained set, carries bound $B_{\mathcal{S}} = 0$ and standard error $\sigma/\sqrt{k}$. The ATT's must bound the full-set violation, $B_{\mathcal{G}} = fV$ with $f = (K - k)/K$, at standard error $\sigma/\sqrt{K}$.

Figure 3 reports the result at $K = 2$, $k = 1$. Panel (a) plots the two honest half-widths. The LATT's is flat in V, since its retained set is clean, while the ATT's rises as it widens to cover the growing violation, the two crossing at $V^* = 1.59\,\sigma$ exactly as Section 4 predicts. Panel (b) is the check that matters. Across the whole range both honest intervals cover the target at 95%, so past V^* the LATT interval is strictly shorter at equal, nominal coverage. The naive ATT interval, which ignores the violation, is narrower still but collapses in coverage, from 95% to below 60%, and is the interval honest inference is meant to replace. Table 4 records the crossing. At $V = 2\sigma$ the LATT half-width $1.96\,\sigma$ undercuts the ATT's $2.16\,\sigma$ with both covering, while the naive interval covers only 71%. Panel (c) traces V^* across designs. In each design the width threshold exceeds the corresponding mean-squared-error threshold, $V/\sigma > \sqrt{1/k - 1/K}\,/f$ with $f = (K - k)/K$, which equals $\sqrt{2}$ at $K = 2$, $k = 1$, since the fixed-length interval charges for the standard-error inflation of dropping cohorts and not only for their variance.

5.6 Selection, calibration, and panel confirmation

Two questions remain for the data-driven regime, whether choosing the selected set from the same data distorts the sensitivity interval, and how to choose M when the selected set, and hence its residual violation, is random. On the first, the cost of choosing the set from the same data is concentrated near the threshold and falls on the variance rather than on gross coverage. The M at which coverage crosses the nominal level is nearly the same for the full-data and sample-splitting procedures ($M \approx 0.048$ and 0.056), so where cohorts are well separated from c selection is effectively deterministic, naive inference suffices, and splitting buys little, consistent with the pointwise reading

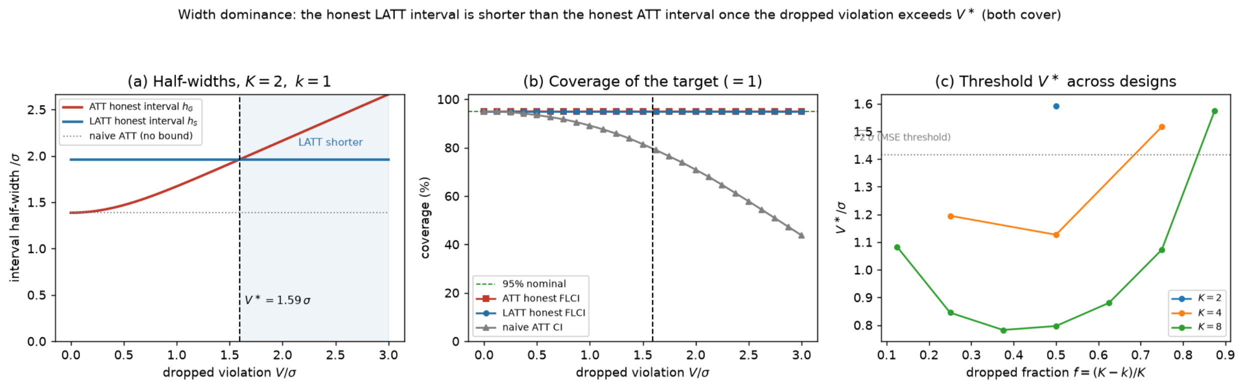

Figure 3: Width dominance (Theorem 2). (a) The honest LATT and ATT half-widths cross at $V^* = 1.59\,\sigma$. Beyond it the LATT interval is shorter. (b) Both honest intervals cover the target at 95% throughout, so the width gain is not bought by undercoverage, while the naive ATT interval collapses. (c) The threshold V^* across designs, above the design-specific mean-squared-error threshold ($\sqrt{2}\,\sigma$ at $K = 2$, $k = 1$).

of Leeb and Pötscher (2005). That near-equality of crossing points understates the cost, however, because selection distorts the standard error rather than the crossing bound, the local-to-threshold regime the carving of Theorem 1 is built for, which the panel model below makes precise. On the second, the residual violation of the selected aggregate is not a constant but a spread, multi-modal random variable across replications, running from zero to about 0.06 with a thin tail to 0.11, each mode a different number of borderline cohorts surviving the screen, which is the content of Remark 3. A fixed M must dominate it, and setting M at the mean residual violation (0.043) undercovers, while nominal coverage is reached near the ninety-fifth percentile (0.057) and well below the maximum (0.112). The lesson is not a value at which to set M, the quantile that would deliver a target coverage depends on the unidentified violation distribution, but a direction in which to read the breakdown curve. A bound drawn from the typical selected set falls at the undercovering end, so the curve should be read conservatively, with M^* weighed against a violation one is willing to countenance rather than against the set's own average.

We next report the carved interval itself, the object Theorem 1 constructs, across the randomization scale γ. In a selection-isolation design where every cohort is clean in the post-period, so the causal target is one for any selected set and the identification widening is held out, cohorts sit near the flatness threshold and the pre- and post-period estimation errors are correlated, so the naive interval on the selected aggregate undercovers. Table 5 traces the sweep. The carved interval covers at the nominal level for every γ, where the naive interval undercovers, confirming the exact conditional coverage of Theorem 1 on the selected set. Its length is summarized by the median rather than the mean, because the carved interval has infinite expected length (Proposition 5): the mean is dominated by the rare near-boundary replications where the truncation collapses, a heavy tail visible in the ninety-fifth percentile, while the median is stable. On the median the carved interval is comparable to a proper $\sqrt{2}$ sample-split, not uniformly shorter, and its median

Table 5: The carved interval across the randomization scale γ (reduced-form model, selection-isolation design, causal target = 1). Coverage is over non-empty selections; length is the median, since the carved interval has infinite expected length (Proposition 5). The carved interval covers where the naive interval undercovers, at a median length comparable to a proper $\sqrt{2}$ sample-split.

	coverage (%)			length			
γ	carved	naive	split	med. carved	med. split	p90 carved	empty (%)
0.0	95.4	90.6	94.6	0.545	0.444	2.47	0.0
0.5	95.8	92.2	95.1	0.555	0.444	2.24	0.0
2.0	95.1	94.1	95.2	0.621	0.512	2.52	0.7
8.0	95.3	94.8	94.5	0.855	0.627	3.17	10.6

length grows with γ as the noised screen shrinks the retained set and the empty-selection event, on which the procedure reports the ATT interval, becomes non-negligible. The carved interval's advantage over splitting is thus not width but that it uses the whole sample deterministically, with the randomization γ trading the length tail against the empty-selection rate; the useful regime is a small γ.

Finally, we repeat the estimand demonstration on generated panel data, where individual outcomes are simulated for a staggered design with a level confound and idiosyncratic noise, cohort event-study coefficients are estimated from genuine difference-in-differences contrasts against a never-treated group, and their covariance is estimated rather than supplied, which is the empirical counterpart of Remark 4. The conclusions are unchanged, since across the informativeness axis the ATT estimator's bias stays flat near +0.30 while the LATT's falls toward zero, reproducing Figure 1 with estimated quantities, and the estimated standard error matches its Monte Carlo counterpart where selection is effectively deterministic. One discrepancy is informative, and it is the payoff promised above. At intermediate informativeness, where selection is stochastic, the local-to-threshold regime the carved interval targets, the estimated standard error understates its Monte Carlo counterpart (0.016 against 0.024 at $\phi = 0.75$), the missing component being the variance induced by the randomness of selection, which a standard error conditional on the realized set does not capture. This confirms, from the panel side, that conditioning on a data-dependent selection understates uncertainty, and why the honest object to report is the sensitivity interval of Section 5.4 rather than a conventional confidence interval.

6 Application: the shale boom and house prices

We illustrate the method with the local effect of the shale boom on house prices, a staggered design with many counties per cohort. The treatment date is the year a county's oil and gas production first rises sharply, from county production records (U.S. Department of Agriculture, Economic Research Service, 2015), which sorts the boom counties into onset cohorts, and the outcome is the log county house-price index (Bogin et al., 2019), over 1998–2019. We estimate Callaway and Sant'Anna (2021) group-time effects against never-treated counties, those with negligible production, screen cohorts on the flatness of their estimated pre-trend, and report the level bound of Section 2 for the selected aggregate.

Onset is gradual and partly endogenous, which is what makes the setting instructive. A county's production ramps up over several years, and where it does so during a regional housing upswing its prices are already climbing before the boom is dated. Figure 4(a) shows the result. The early cohorts have flat pre-trends and are retained, while the later cohorts, whose onset coincides with the mid-2000s house-price run-up, carry pre-treatment trends several times larger and are dropped.

The two aggregates part ways (Figure 4b). Pooled across all cohorts, the event study climbs through the onset date to a house-price effect of +6.7 log points ($t = 8$), the kind of estimate a practitioner would report as a clear positive effect of the boom. But the pooled path is already trending before treatment, and the effect is inherited from the dropped cohorts. Restricted to the credible subpopulation, the three cohorts whose pre-trends are essentially flat (the screen at $c \approx 0.03$), the event study is flat both before and after onset, and the LATT is -0.2 log points, indistinguishable from zero. The gap between the two, +6.9 log points, is itself sharply estimated ($t = 5$). Honest inference confirms the reading rather than resting on the point estimate. Applying the level bound of Section 2 to the credible aggregate at the illustrative value M equal to the screen tolerance $c \approx 0.03$, one point on the breakdown curve, the LATT's fixed-length interval runs from -5.9 to $+5.5$ log points, which contains zero and excludes the pooled +6.7. The choice $M = c$ is a benchmark rather than a consequence of the screen, since c bounds the pre-treatment coefficients while M bounds the unobserved post-treatment violation, so the interval is honest only under the maintained level bound at that M. The breakdown value is $M^* \approx 4.2$ log points, so reconciling the credible near-zero reading with the pooled effect would require countenancing a post-treatment violation about 1.4 times the screen tolerance and larger than any retained cohort's own pre-trend. Panel (c) traces the estimate along the credibility path of Section 2.3. It stays flat and close to zero across the whole range of genuinely flat pre-trends, cohorts with $\max_e |\hat{\beta}_{g,\text{pre}}(e)|$ below about 0.03, and climbs toward the pooled +6.7 only once the threshold is loosened enough to readmit cohorts whose own pre-trends, around 0.07–0.08, are as large as the effect in question. Reading the LATT at that lax end simply rebuilds the pooled estimate out of its least credible cohorts, and the positive figure survives only by abandoning the credibility standard, not by tolerating a residual violation of the credible set.

Because the credible set is read from the same estimated pre-trends, we run the carved procedure of Theorem 1 on the data, estimating the covariance of the stacked cohort event-study coefficients by a cluster bootstrap over counties, the estimated-Σ case of Remark 4. The screen boundary is genuinely uncertain here, which is what makes carving operative rather than cosmetic. Only the 2003 cohort is retained by a wide margin, and 2007, 2008, and 2011 are dropped by one, while four of the nine cohorts sit within about one bootstrap standard error of the threshold, 2004 and 2005 retained and 2006 and 2009 dropped. Carrying that selection uncertainty, the carved interval for the credible LATT at $\gamma = 0$ is $[-4.3, +3.3]$ log points, tracking the naive interval because the retained aggregate stays near zero across the plausible selected sets. Composing it with the level bound at $M = c$, the fully honest data-driven object of Corollary 1, gives $[-7.3, +6.3]$ log points, which still contains zero with its upper edge below the pooled $+6.7$; the near-optimal fixed-length interval $[-5.9, +5.5]$ is its counterpart valid under separation (Remark 1). The carved machinery thus confirms that the near-zero credible reading is robust to the selection uncertainty the pre-trend screen introduces, and not an artifact of conditioning on the realized selected set.

We checked this against the sensitivity analysis a fixed-target approach would actually run, the HonestDiD implementation of Rambachan and Roth (2023) applied to the pooled event study. We use their relative-magnitudes restriction $\Delta^{RM}(\bar{M})$, the "post bounded by pre" restriction, which bounds post-treatment trend changes by $\bar{M}$ times the largest pre-treatment one, because it reads robustness directly off the observed pre-trend, the mechanism at issue here. The honest ATT confidence set is $[+1.0, +13.3]$ log points at $\bar{M} = 1$, wider than the naive interval $[+5.1, +8.4]$, but still excluding zero, and first admits zero only at a breakdown of $\bar{M}^* = 1.2$. The relative-magnitudes restriction thus reports the $+6.7$ effect as robust, surviving post-treatment violations larger than the entire visible pre-trend. That robustness is an artifact of aggregation. The pooled pre-trend is nearly flat, its largest pre-treatment coefficient is under one log point, because the confounded cohorts are a minority whose large, differently-timed pre-trends wash out in the all-cohort average, even as their own pre-trends reach eleven log points. Reading only the aggregate, the restriction cannot detect this and certifies the effect, the false precision that Section 5.4 anticipated for a bound read off an uninformative pre-trend, here on real data.

The complementary smoothness restriction $\Delta^{SD}(M)$ fails in the opposite direction, for a reason specific to this confound. Run on the same pooled event study through the same conditional–hybrid procedure, its honest confidence set admits zero once the post-treatment differential trend is allowed to curve by $M \approx 0.001$, a fifth of the ≈ 0.005 curvature the pooled pre-trend itself displays, so it neither certifies the $+6.7$ effect nor localizes it, but widens to include both zero and the full effect.[3] The housing bubble is a smooth secular appreciation, and a smoothness restriction cannot separate it from a treatment effect on the aggregate. The two standard restrictions thus fail in opposite ways, relative magnitudes falsely precise, smoothness uninformative, and neither recovers the near-zero reading the cohort-level screen delivers.

The credible-subpopulation screen, reading pre-trends cohort by cohort, drops exactly the bubble cohorts and returns the near-zero LATT. It is the only one of the three analyses that both shows no effect for the credible cohorts and locates the pooled positive in the dropped ones, because the information distinguishing bubble from effect lives in the cross-cohort composition that every aggregate analysis first averages away. Whether the dropped cohorts' excess is differential trend or genuine heterogeneous effect is the unidentified composition gap, so the analysis concentrates the pooled positive in cohorts with nonflat pre-trends rather than proving the full-population effect zero.

The report of Section 4 makes this quantitative. The estimable divergence between the two

[3] We report Δ^{SD} through the same conditional least-favourable hybrid used for Δ^{RM}, so the two restrictions are compared under one inference procedure. The smoothness class also admits the near-optimal fixed-length interval of Armstrong and Kolesár (2018), which is tighter but leaves the comparison unchanged. Near the breakdown the identified set is small and the hybrid is close to optimal.

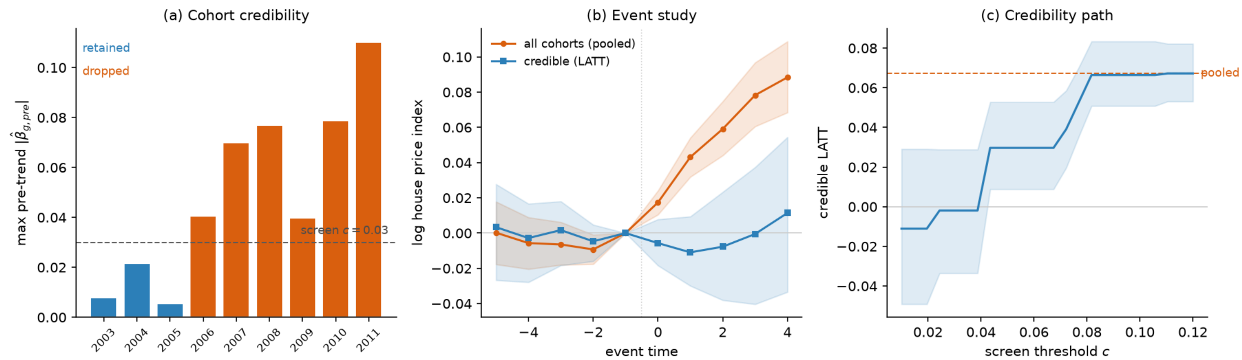

Figure 4: The shale boom and county house prices. (a) Cohort credibility map, the maximum pre-treatment violation by onset cohort with the flatness screen, where the later, trending cohorts are dropped. (b) Event study, all cohorts (pooled estimate) versus the credible subpopulation (LATT) at the illustrative threshold $c \approx 0.03$, with 95% bands. (c) The credibility path, the LATT as the screen threshold c is relaxed, near zero across the flat-pre-trend cohorts and rising toward the pooled estimate only as trending cohorts are readmitted.

point estimators is $\hat{D} = \hat{\theta}_{\mathcal{G}} - \hat{\theta}_{\hat{S}} = +6.9$ log points ($t = 5.4$, carved standard error), and it splits as $\hat{D} = B_{\mathcal{F}} - \Gamma$ into the dropped cohorts' differential trend $B_{\mathcal{F}}$, a pre-trend violation, and the composition gap Γ, a difference in true effects, which the data cannot separate. The breakdown value is $\Gamma^* \approx -3.3$ log points, so the credible reading is overturned only if more than about half of the divergence, 3.3 of the 6.9 log points, reflects genuine treatment-effect heterogeneity rather than differential pre-trend. Reporting both honest intervals with Γ^* completes the standing rule; the width comparison of the switching rule is an efficiency question, demonstrated in Section 5.5, and here the two honest intervals disagree on substance rather than width.

The reading is that the pooled effect reflects a pre-existing trend in the counties that adopt late rather than a credibly causal effect, and a near-zero net effect for the credible subpopulation is what the offsetting amenity and disamenity channels of local energy development would predict (Muehlenbachs et al., 2015). The method identifies the effect for the credible cohorts alone, and the excess in the dropped cohorts blends their steep pre-trends with any genuine difference in their treatment effects, a mixture the design cannot separate and need not, since the credible reading stands on the retained cohorts. The application thus shows the method working in the direction opposite to point-identification, not recovering an effect the pooled estimate misses but withdrawing one it spuriously reports, and locating that withdrawal exactly in the cohorts the screen flags.

7 Conclusion

We have argued that when parallel trends fails for some treated cohorts, a productive response is to change the estimand rather than to defend or bound the ATT. The credible-subpopulation LATT is point-identified under parallel trends for the selected cohorts alone, a condition that can hold when the ATT's fails, is estimated by a transparent reweighting of standard group-time effects, and can be accompanied by honest sensitivity bounds on the residual violation that the pre-trend screen cannot rule out. This is a specific instance of the general principle, familiar from instrumental variables and limited overlap, of retreating to a credibly-identified subpopulation.

The method delivers a sharp answer where the ATT estimator is biased and a fixed-target sensitivity analysis on the aggregate is uninformative, wide when the aggregate pre-trend reveals the violation, and falsely precise when the offending cohorts wash out of it, at the price of a narrower, subpopulation target whose composition must be described rather than assumed. Its advantage is contingent on pre-trends being informative about post-treatment violations. Where they are not, only economic context can carry the assumption. Under data-driven selection the estimator carries a pre-test bias that vanishes as pre-trends grow informative. The inference we recommend is honest regardless, combining randomized selective inference (Fithian et al., 2014; Tian and Taylor, 2018) with the sensitivity bounds of Rambachan and Roth (2023) into an interval uniformly valid against both the selection and the residual violation. We also make the scope condition operational. A feasible rule, reported alongside both honest intervals, signals when narrowing the target lowers risk, with a breakdown value Γ^* standing in for the one quantity, the effects of the discarded cohorts, that the data cannot identify.

An appendix characterizes the optimal credible weighting (Proposition 7). Minimizing worst-case risk yields a soft-thresholded weighting of cohorts by credibility net of noise, of which the flatness screen with size weights is a special case, which completes the analogy to the variance-minimizing overlap weights of Crump et al. (2009) that the approach borrows from.

Two directions remain for future work. One is the finite-sample behaviour of the feasible optimal weighting of Appendix A, where the credibility levels m_g are estimated from the pre-trends rather than known, and the noise this inherits awaits a full account. The other is the coverage of a single interval selected by a hard-thresholded switching rule. The recommended report of both intervals is honest whatever the rule decides (Proposition 6), so the question is confined to the coverage of one interval at the switching boundary.

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A Optimal credible weighting

The flatness screen and the cohort weights w_g have so far been taken as given, the screen retaining a cohort outright and the weights fixed at, say, cohort size. Both are choices, and the sensitivity class that governs the honest inference also pins down their optimal form, in the decision-theoretic sense in which Crump et al. (2009) derive the optimal-overlap subpopulation and Li et al. (2018) the overlap weights. There, variance is minimized where positivity is scarce. Here, worst-case mean squared error is minimized where credibility is scarce.

Let the estimator be a weighted average $\hat{\theta}(\lambda) = \sum_g \lambda_g \hat{\beta}_{g,\text{post}}$ with $\lambda_g \geq 0$ and $\sum_g \lambda_g = 1$, and, following the overlap-weighting logic that the weights define the estimand as well as the estimator, let the target be the λ-weighted causal effect $\theta(\lambda) = \sum_g \lambda_g \theta_g$, whose interpretation is reported through the weights themselves. Since $\hat{\beta}_{g,\text{post}} \sim \mathcal{N}(\theta_g + V_g, \sigma_g^2)$ independently, the estimator has identification bias $\sum_g \lambda_g V_g$ and variance $\sum_g \lambda_g^2 \sigma_g^2$. The pre-trends enter through the sensitivity class. A cohort's post-treatment violation is bounded by a credibility level $m_g \geq 0$, $|V_g| \leq m_g$, with m_g small for a cohort whose pre-trend is flat and large for one whose pre-trend is steep. The hard flatness screen is the coarsening $m_g \in \{0, \infty\}$, retaining the $m_g = 0$ cohorts and discarding the rest. The level bound M of Section 3 is the common value $m_g \equiv M$ on the retained set. The worst-case mean squared error over this class is

$$\mathcal{R}(\lambda) = \Big(\sum_g \lambda_g m_g\Big)^2 + \sum_g \lambda_g^2 \sigma_g^2,$$

the squared worst-case bias plus the variance, and the optimal credible weighting minimizes it over the simplex.

Proposition 7 (Optimal credible weighting)**.** *$\mathcal{R}$ is strictly convex on the simplex, so it has a unique minimizer λ^*. There are scalars μ and $B = \sum_g \lambda_g^* m_g$, the worst-case bias at the optimum, such that*

$$\lambda_g^* = \frac{1}{\sigma_g^2}\Big[\tfrac{\mu}{2} - B\, m_g\Big]_+, \qquad [x]_+ = \max(x, 0),$$

with μ fixed by $\sum_g \lambda_g^* = 1$. When $B > 0$ there is an endogenous credibility cutoff $m^* = \mu/(2B)$. Cohorts with $m_g \geq m^*$ receive zero weight, and among the retained cohorts the weight is $\lambda_g^* \propto (m^* - m_g)/\sigma_g^2$, decreasing in the credibility bound m_g and in the sampling variance σ_g^2. When $B = 0$, as under the benchmark $m_g \equiv 0$, the cutoff is vacuous and $\lambda_g^* \propto 1/\sigma_g^2$, inverse-variance weights with no thresholding.

Proof. $\sum_g \lambda_g^2 \sigma_g^2$ is strictly convex for $\sigma_g^2 > 0$ and $(\sum_g \lambda_g m_g)^2$ is convex, so $\mathcal{R}$ is strictly convex and its minimizer over the compact convex simplex is unique. The worst-case bias is $\sup_{|V_g| \leq m_g} |\sum_g \lambda_g V_g| = \sum_g \lambda_g m_g$ since $\lambda_g \geq 0$, giving the displayed $\mathcal{R}$. Form the Lagrangian $\mathcal{R}(\lambda) - \mu(\sum_g \lambda_g - 1) - \sum_g \nu_g \lambda_g$ with $\nu_g \geq 0$. Stationarity gives $2Bm_g + 2\lambda_g \sigma_g^2 - \mu - \nu_g = 0$ with $B = \sum_g \lambda_g m_g$. For $\lambda_g > 0$ complementary slackness sets $\nu_g = 0$ and $\lambda_g = (\mu/2 - Bm_g)/\sigma_g^2$. For $\lambda_g = 0$ it requires $\mu/2 - Bm_g \leq 0$, i.e. $m_g \geq \mu/(2B)$. Combining the two cases gives the truncated form and the cutoff $m^* = \mu/(2B)$, with μ determined by the budget constraint and B by its own definition at the resulting λ^*. $\square$

Three readings connect the result to the rest of the paper. First, the optimal rule is a soft flatness screen. It reproduces the hard screen's qualitative content, cohorts whose pre-trends are too steep are dropped, but with an endogenous threshold set by the bias–variance trade-off rather than a researcher's c, and with the surviving cohorts weighted continuously by how credible and how precise they are, rather than equally. A marginally less credible but large and precisely estimated cohort is retained where the hard screen would discard it, because the variance its inclusion saves outweighs the bias it admits, the same calculus as the dominance condition of Proposition 8, now internal to the weighting. Second, the result justifies the paper's default. Among equally credible cohorts, $m_g \equiv 0$, the weights collapse to inverse-variance, $\lambda_g^* \propto 1/\sigma_g^2$, and when sampling variance scales inversely with cohort size, $\sigma_g^2 \propto 1/n_g$, inverse-variance weighting is size weighting. The fixed w_g of the definition is thus optimal precisely under homoskedastic-per-unit noise and equal credibility, and the proposition says what to do when either fails. Third, it completes the analogy to limited overlap. Crump et al. (2009) trim and Li et al. (2018) weight to minimize variance where positivity is the binding scarcity. Here credibility is the scarcity, and the optimal response is to weight cohorts by credibility net of noise, with trimming emerging as the corner solution for cohorts past m^*.

The credibility levels m_g are reported, not assumed, in the same currency as everything else. A natural choice sets m_g to the cohort's own maximal pre-trend, $m_g = \max_{e<0}|\hat{\beta}_{g,\text{pre}}(e)|$, so that the weighting is read off the observed pre-trends. This makes the weights data-driven, a refinement of the selection already treated, and, as there, validity does not rest on it. The honest interval of Section 3 takes its coverage from the researcher-set level bound M, while the estimated m_g only orders and weights the cohorts. Using m_g to weight but M to cover keeps efficiency and honesty on separate instruments, as the carved inference of Theorem 1 keeps selection and identification on separate instruments. One caveat separates the feasible rule from Proposition 7. That optimality is minimax over the class $|V_g| \leq m_g$, so it holds for the feasible weights only if the observed pre-trend m_g bounds the post-treatment violation V_g. Absent such a pre-to-post link, the feasible m_g is a credibility ranking rather than the bound defining the optimized class, and the weighting is a heuristic that need not inherit the minimax interpretation. A second caveat concerns coverage. Because the estimated weights make the contrast ℓ_S and the causal target random functions of the pre-trends, the discrete-selection guarantee of Theorem 1, which conditions on a fixed contrast, does not extend to the continuously weighted interval. The level bound M absorbs the residual identification bias, but the sampling and selection uncertainty from estimating the weights requires a separate argument, which we leave to future work. A practitioner wanting the feasible weighting with valid inference today can estimate the weights on an independent split and invert on the other, at the usual efficiency cost; honest inference that carves the continuous weighting without that split is the open question.

Table 6: Optimal weighting versus the hard screen (reduced-form model, homogeneous effects, so all procedures target the same value). RMSE and MSE for that target. The optimal weighting minimizes worst-case MSE by Proposition 7.

Procedure	RMSE	MSE
ATT (all cohorts)	0.200	0.0402
Hard screen (best c)	0.032	0.00104
Optimal weighting, feasible ($\hat{m}_g$)	0.022	0.00048
Optimal weighting, oracle ($m_g = V_g$)	0.020	0.00039

A.1 The optimal weighting improves on the hard screen

Proposition 7 makes the soft weighting weakly dominant over the hard screen by construction, the screen being the coarsening $m_g \in \{0, \infty\}$ of the same objective, so the only empirical question is how large the gain is. We measure it in a homogeneous-effects design ($K = 12$ cohorts in precise/noisy pairs across six violation levels $V_g \in \{0, 0.08, \ldots, 0.40\}$, so every weighting targets the same value and mean squared error is comparable), against the ATT, the best-threshold hard screen, and the optimal weighting in an oracle form ($m_g = V_g$ known) and a feasible form ($\hat{m}_g = \max_e |\hat{\beta}_{g,\text{pre}}(e)|$ estimated at informativeness $\phi = 1.5$). Table 6 reports the result. Against the best tabulated hard screen the oracle weighting lowers MSE by 63% and the feasible weighting by 54%; against a feasible hard screen that also estimates $\hat{m}_g$ the reduction is 28%, and across 500 random DGPs the typical gain is 14%. The mechanism is the precision channel the proposition predicts: at a common credibility level the optimal weighting loads on the precisely estimated cohort and soft-thresholds the credibility gradient where the hard screen weights retained cohorts equally. The gap between the oracle and feasible numbers is the price of estimating m_g from noisy pre-trends, the open question the conclusion notes.

B Mean-squared-error dominance

This appendix develops the point-estimator account behind the width comparison of Section 4, identifying which three forces set the sign of the advantage and the explicit thresholds they imply. Changing the estimand is worth doing only when the bias it removes outweighs the variance and the change of target it costs. Three forces set the sign, the violation the screen removes, the variance that dropping a cohort costs, and the heterogeneity between the retained subpopulation and the full one, and the sign turns on their balance, not on the informativeness of the pre-trends, which governs only how much of a fixed-sign advantage is realized.

We state the comparison for K cohorts, and then read from the two-cohort special case the one thing the general result leaves implicit, how the advantage scales with the informativeness of the pre-trends. Let each cohort g carry a weight w_g (normalized so $\sum_g w_g = 1$), a causal effect $\theta_g = \text{ATT}(g, \cdot)$, a post-treatment violation V_g (the aggregate $a(\delta_{g,\text{post}})$, zero for a clean cohort), and an independent sampling variance σ_g^2 for its aggregated post-treatment estimate. The screen retains cohort g with probability π_g, independently across cohorts and of the post-treatment estimates. The ATT estimator is the fixed average $\sum_g w_g \hat{\beta}_{g,\text{post}}$. The LATT estimator reweights over the retained set $\hat{S}$, $\sum_{g \in \hat{S}} w_g \hat{\beta}_{g,\text{post}} / W_{\hat{S}}$ with $W_S = \sum_{g \in S} w_g$.

Proposition 8 (General-K dominance)**.** *With the target $\theta_{\mathcal{G}} = \sum_g w_g \theta_g$, the ATT estimator has mean squared error $\mathrm{MSE}_{\mathrm{ATT}} = \left(\sum_g w_g V_g\right)^2 + \sum_g w_g^2 \sigma_g^2$, and the LATT estimator has*

$$\mathrm{MSE}_{\mathrm{LATT}} = \sum_{S \subseteq \mathcal{G}} \Big(\textstyle\prod_{g \in S} \pi_g \prod_{g \notin S} (1 - \pi_g) \Big) \Big[\big(\theta_S - \theta_{\mathcal{G}} + B_S\big)^2 + \mathrm{Var}_S \Big], \tag{6}$$

where $\theta_S = \sum_{g \in S} w_g \theta_g / W_S$, $B_S = \sum_{g \in S} w_g V_g / W_S$, and $\mathrm{Var}_S = \sum_{g \in S} w_g^2 \sigma_g^2 / W_S^2$, with the convention $S = \mathcal{G}$ when the screen retains no cohort. When every cohort is retained, $\pi_g = 1$, the two estimators coincide and $\mathcal{A} = \mathrm{MSE}_{\mathrm{ATT}} - \mathrm{MSE}_{\mathrm{LATT}} = 0$. Suppose instead the screen separates the cohorts, discarding every confounded cohort, $\pi_g \to 0$ for $V_g \neq 0$, and retaining every clean cohort, $\pi_g \to 1$ for $V_g = 0$, so that the retained set converges in probability to the clean set $\mathcal{C}$ and the dropped set is the confounded set $\mathcal{F} = \mathcal{G} \setminus \mathcal{C}$. Then

$$\mathcal{A} \;\longrightarrow\; \mathcal{A}^* \;=\; B_{\mathcal{F}}^2 - \Gamma^2 - \Delta\mathrm{Var}, \tag{7}$$

where $B_{\mathcal{F}} = \sum_{g \in \mathcal{F}} w_g V_g$ is the aggregate violation removed by selection (a realized quantity, unlike the worst-case level bound $B_{\hat{S}}$ of Corollary 1 that widens the honest interval), $\Gamma = \theta_{\mathcal{C}} - \theta_{\mathcal{G}}$ is the composition gap between the retained subpopulation and the full population, and $\Delta\mathrm{Var} = \mathrm{Var}_{\mathcal{C}} - \sum_g w_g^2 \sigma_g^2 \geq 0$ is the variance cost of dropping. Hence in the informative limit the LATT dominates the ATT in mean squared error if and only if

$$B_{\mathcal{F}}^2 \;>\; \Gamma^2 + \Delta\mathrm{Var}. \tag{8}$$

With two equally weighted cohorts, one clean and one confounded with violation V, effect gap γ, and common variance σ^2, one has $B_{\mathcal{F}}^2 = V^2/4$, $\Gamma^2 = \gamma^2/4$, and $\Delta\mathrm{Var} = \sigma^2/2$, so (8) *becomes $V^2 > \gamma^2 + 2\sigma^2$. Remark 5 below develops this two-cohort case, where the informativeness of the pre-trends enters.*

Proof. See Appendix C. □

The general threshold (8) is the two-cohort comparison written for many cohorts, each of its three forces now an aggregate. The reason to trade is the total violation $B_{\mathcal{F}}$ selection removes, not any single cohort's. The estimand cost is the composition gap Γ between the retained subpopulation's effect and the full population's, the many-cohort form of γ. It vanishes under the equal-targets condition, that the retained and dropped cohorts share the same average effect. The variance cost $\Delta\mathrm{Var}$ is the extra sampling variance from reweighting onto fewer cohorts. The sign of the advantage is a violation-against-noise-and-heterogeneity question, settled by (8) at the endpoints where the screen retains everyone ($\mathcal{A} = 0$) or fully separates the cohorts ($\mathcal{A} = \mathcal{A}^*$). Between them the path need not be monotone. With more than two cohorts, partial selection can retain a subset of the confounded cohorts, and when their violations cancel in the full ATT this can raise the LATT's bias, sending $\mathcal{A}$ negative before it recovers to $\mathcal{A}^*$ at separation, so the monotone reading of $\mathcal{A}$ running from zero to $\mathcal{A}^*$ as informativeness grows is specific to the two-type clean-versus-confounded case. Two qualifications connect this to the inference in the main text. The separation that makes (7) exact is the mean-squared-error counterpart of the condition behind Remark 1. Near-threshold cohorts are retained with interior probability, the same local-to-threshold configurations that obstruct uniform post-selection inference. And an undetectable confounded cohort, flat before treatment yet violating parallel trends after, stays in both estimators, so it enters (7) through the residual of Proposition 3 that the honest inference of Section 3 bounds.

The two-cohort case makes the three forces explicit and, uniquely, traces the advantage along the informativeness axis. Consider two equally weighted cohorts sharing a common sampling variance σ^2, observed as one pre- and one post-treatment coefficient each, with independent pre- and post-period sampling errors. One cohort is clean, $\delta_1 = 0$. The other is confounded, with a post-treatment violation $V > 0$ and a pre-treatment violation ϕV, where $\phi \geq 0$ is the informativeness of the pre-trend. Their treatment effects are τ_1 and τ_2, with $\gamma := \tau_2 - \tau_1$, so the target is the average $\bar{\tau} = (\tau_1 + \tau_2)/2$. Taking c large enough that the clean cohort is retained with probability approaching one, the confounded cohort survives the screen $|\hat{\beta}_{2,\mathrm{pre}}| \leq c$ with probability

$$p_2(\phi) \;=\; \Phi\Big(\tfrac{c-\phi V}{\sigma}\Big) - \Phi\Big(\tfrac{-c-\phi V}{\sigma}\Big), \tag{9}$$

which decreases from near one, when the confound is invisible in the pre-period, to zero as the pre-trend announces it.

Remark 5 (Informativeness scaling, two cohorts). In the two-cohort model, with the clean cohort retained with probability approaching one, the mean-squared-error advantage of the LATT estimator over the ATT estimator is

$$\mathcal{A}(\phi) \;=\; \mathrm{MSE}_{\mathrm{ATT}} - \mathrm{MSE}_{\mathrm{LATT}} \;=\; \frac{1 - p_2(\phi)}{4}\left[V^2 - \gamma^2 - 2\sigma^2\right], \tag{10}$$

the general advantage (7) specialized by $B_{\mathcal{F}}^2 = V^2/4$, $\Gamma^2 = \gamma^2/4$, $\Delta\mathrm{Var} = \sigma^2/2$ and scaled by the survival factor $1 - p_2(\phi)$ (derivation in Appendix C). The bracket's three terms are the forces of (8) in closed form, $+V^2$ the bias the screen removes, $-2\sigma^2$ the variance that dropping a cohort costs, and $-\gamma^2$ the estimand-change cost, charged only to a reader who insists on the ATT as the common target and vanishing under homogeneity $\gamma = 0$. Hence the LATT dominates the ATT in mean squared error if and only if

$$V^2 \;>\; \gamma^2 + 2\sigma^2, \tag{11}$$

a sign fixed independently of the informativeness ϕ, which enters only through the nondecreasing factor $1 - p_2(\phi)$. Two readings follow. Informative pre-trends are necessary but not sufficient. If $V^2 < \gamma^2 + 2\sigma^2$ no degree of informativeness rescues the LATT, since $1 - p_2$ only drives an already-negative bracket further from zero, informativeness scales the advantage, it does not create one. And when the bracket is positive the advantage rises monotonically in informativeness, from zero at $\phi = 0$ to its ceiling $\frac{1}{4}[V^2 - \gamma^2 - 2\sigma^2]$ as $\phi \to \infty$, the vertical gap between the two bias curves in Figure 1.

The dominance condition can be read off the data, up to the one quantity it cannot identify, and so furnishes the diagnostic reported in the switching rule (Proposition 6). The variance cost $\Delta\mathrm{Var}$ is a function of Σ and the weights alone and is therefore estimable, and the difference between the two point estimators consistently estimates the violation net of the composition gap,

$$\hat{\theta}_{\mathcal{G}} - \hat{\theta}_{\hat{S}} \xrightarrow{p} B_{\mathcal{F}} - \Gamma \;=:\; D, \tag{12}$$

since the ATT estimator carries the dropped cohorts' violation while the LATT estimator does not. What is not identified is Γ on its own, because it depends on the counterfactual effects of the very cohorts the screen discards. Writing $B_{\mathcal{F}} = D + \Gamma$ in (8), the advantage $\mathcal{A}^* = B_{\mathcal{F}}^2 - \Gamma^2 - \Delta\mathrm{Var}$ becomes $D^2 + 2D\Gamma - \Delta\mathrm{Var}$, linear in the unidentified Γ because the squared composition gap cancels. Two readings follow. Under homogeneous effects, $\Gamma = 0$, the trade pays exactly when the squared gap between the estimators exceeds the variance cost,

$$D^2 \;>\; \Delta\mathrm{Var}, \tag{13}$$

a comparison the practitioner can compute and, using the carved standard error of Theorem 1 for D, test. In general the verdict is monotone in Γ, increasing when $D > 0$, and flips at the breakdown value $\Gamma^* = (\Delta\mathrm{Var} - D^2)/(2D)$, the quantity Proposition 6(iii) reports. The trade pays unless the dropped cohorts' effects differ from the retained ones by more than Γ^* allows, a sensitivity statement in the same idiom as the breakdown value M^* for the identifying restriction, now for the estimand change rather than the violation.

Three qualifications sharpen the dominance calculation. The advantage (10) is an upper bound on what the method delivers, because it omits two channels that both work against the LATT, occasional false rejection of the clean cohort, which enters at order $1 - p_1$ when the threshold is not comfortably large, and the additional variance that data-driven selection induces and that a standard error conditional on the realized set does not capture, the understatement documented in Section 5.6. Both shrink the realized advantage toward zero, and near the threshold, where the dominance margin $V^2 - \gamma^2 - 2\sigma^2$ is small, a positive omitted cost can reverse the ranking, so the sign is guaranteed only when that margin exceeds these costs. Finally, (10) is a comparison of point estimators under squared-error loss, not of the honest intervals that are the paper's recommended output. The interval analogue replaces the reason-to-trade V with the residual violation net of the calibration buffer of Remark 3, and the variance cost σ^2 with a sampling slack inflated by the selection quantile, so that the level-bound half-width comparison turns on the same three forces with V discounted and σ inflated. The mean-squared-error threshold (11) correctly signs that comparison and is the transparent summary. The width comparison of Section 4 sharpens the constants.

C Omitted proofs

Proof of Proposition 1. For $g \in S$, $\beta_{g,\text{post}} = \tau_{g,\text{post}} + \delta_{g,\text{post}} = \tau_{g,\text{post}}$ by the hypothesis and (2), so $a(\beta_{g,\text{post}}) = a(\tau_{g,\text{post}}) = \text{ATT}(g,\cdot)$ by linearity of a. The reweighted sum over S therefore equals the weighted average of the aggregated causal effects, which is θ_S by definition. Cohorts outside S receive zero weight, so their differential trends do not enter. $\square$

Proof of Corollary 1. Theorem 1 gives $\theta_{\hat{S}} \in \mathcal{C}_{1-\alpha}(\hat{S})$ with probability $1-\alpha$. The maintained restriction places $\theta^{\text{c}}_{\hat{S}}$ within $B_{\hat{S}}$ of $\theta_{\hat{S}}$ deterministically, so $\{\theta_{\hat{S}} \in \mathcal{C}_{1-\alpha}(\hat{S})\} \subseteq \{\theta^{\text{c}}_{\hat{S}} \in \mathcal{C}^{M}_{1-\alpha}(\hat{S})\}$, and monotonicity of probability gives the bound. $\square$

Proof of Proposition 5. By Theorem 1, conditional on $\{\hat{S} = S\}$ and the orthogonal remainder r, $\hat{\theta}_S \sim \mathcal{TN}(\theta_S, s_S^2, [\mathcal{V}^-, \mathcal{V}^+])$, and $\mathcal{C}_{1-\alpha}(S)$ inverts the truncated-normal pivot in θ_S, which is exactly the confidence set of Lee et al. (2016) for a normal mean under a polyhedral, hence after conditioning on r interval, truncation. Kivaranovic and Leeb (2021) show this set has infinite expected length whenever the truncation is one-sided with positive probability. Under the flatness screen a retained cohort's binding constraint is two-sided only when both faces $|\hat{\beta}_{g,\text{pre}}(e) + \omega_g| \le c$ are active; generically one face binds and the induced bound on $\hat{\theta}_S$ is one-sided, an event of positive probability, so the hypothesis holds. Averaging over S preserves the conclusion. That only $\gamma \to \infty$ escapes is immediate, since it empties the selection. $\square$

Proof of Proposition 6. (i) Each interval's coverage is a property of that interval alone and holds for every β. Reporting both leaves each untouched, so whichever the rule recommends, the reader is handed an interval that covers at $1-\alpha$. (ii) By Theorem 2, $h_S < h_{\mathcal{G}} \iff B_{\mathcal{G}} > B^*_{\mathcal{G}}$, and both half-widths are functions of $\hat{\Sigma}$ and the chosen bounds. For a fixed retained set $\hat{D} \xrightarrow{p} -\Gamma + (\delta_{\mathcal{G}} - \delta_S)$, which equals $B_{\mathcal{F}} - \Gamma$ in the informative limit where the retained aggregate violation vanishes and the full aggregate violation is carried by the dropped cohorts. Its standard error is that of the contrast $(\ell_{\mathcal{G}} - \ell_S)'\hat{\beta}_{\text{post}}$, carved for the selection. (iii) The composition gap $\Gamma = \theta_{\hat{S}} - \theta_{\mathcal{G}}$ depends only on the discarded cohorts' effects and is therefore unidentified. The explicit Γ^* follows from the mean-squared-error advantage of Appendix B. $\square$

Proof of Proposition 8. The ATT estimator is a fixed linear combination with mean $\theta_{\mathcal{G}} + \sum_g w_g V_g$ and variance $\sum_g w_g^2 \sigma_g^2$, giving MSE_{ATT}. For the LATT, condition on the retention configuration. On the event that the retained set is S, which has probability $\prod_{g\in S} \pi_g \prod_{g \notin S}(1-\pi_g)$, the estimator is $\sum_{g\in S} w_g \hat{\beta}_{g,\text{post}}/W_S$, whose mean is $\theta_S + B_S$ and whose variance is Var_S, since retention is independent of the post-treatment estimates. Its error relative to $\theta_{\mathcal{G}}$ has squared bias $(\theta_S - \theta_{\mathcal{G}} + B_S)^2$ and variance Var_S. Averaging over configurations gives (6). At $\pi_g = 1$ only $S = \mathcal{G}$ has mass, where $\theta_{\mathcal{G}} - \theta_{\mathcal{G}} + B_{\mathcal{G}} = \sum_g w_g V_g$ and $\text{Var}_{\mathcal{G}} = \sum_g w_g^2 \sigma_g^2$, so $\text{MSE}_{\text{LATT}} = \text{MSE}_{\text{ATT}}$ and $\mathcal{A} = 0$. Under separation the configuration $S = \mathcal{C}$ carries probability approaching one, so $\text{MSE}_{\text{LATT}} \to (\theta_{\mathcal{C}} - \theta_{\mathcal{G}})^2 + \text{Var}_{\mathcal{C}}$, using $B_{\mathcal{C}} = 0$ because clean cohorts have $V_g = 0$. Since clean cohorts contribute nothing to $\sum_g w_g V_g$, that sum equals $B_{\mathcal{F}}$, and $\mathcal{A} \to B^2_{\mathcal{F}} + \sum_g w_g^2 \sigma_g^2 - \Gamma^2 - \text{Var}_{\mathcal{C}} = B^2_{\mathcal{F}} - \Gamma^2 - \Delta\text{Var}$, which is (7). The threshold (8) is its positivity, and the two-cohort reduction is the displayed substitution. $\square$

Proof of Remark 5. Write $\tau_1 = \bar{\tau} - \gamma/2$, $\tau_2 = \bar{\tau} + \gamma/2$, so $\hat{\beta}_{1,\text{post}} \sim \mathcal{N}(\bar{\tau} - \gamma/2, \sigma^2)$ and $\hat{\beta}_{2,\text{post}} \sim \mathcal{N}(\bar{\tau} + \gamma/2 + V, \sigma^2)$, independent of each other and of the retention indicators. The ATT estimator has $\text{MSE}_{\text{ATT}} = V^2/4 + \sigma^2/2$. With the clean cohort always retained, the LATT estimator is the two-cohort average with probability p_2 and $\hat{\beta}_{1,\text{post}}$ with probability $1 - p_2$, so $\text{MSE}_{\text{LATT}} = p_2(V^2/4 + \sigma^2/2) + (1 - p_2)(\gamma^2/4 + \sigma^2)$. Subtracting gives (10), which is (7) under the stated substitution. Since $1 - p_2 \geq 0$ the sign is that of the bracket, giving (11), and $p_2(\phi)$ is nonincreasing in ϕ for $\phi V \geq 0$, so $1 - p_2$ is nondecreasing. $\square$

Proof of Theorem 2. By (5) each interval is the near-optimal fixed-length interval of Armstrong and Kolesár (2018), with half-width $h(B, s) = s\,\text{cv}_\alpha(B/s)$ and coverage at least $1-\alpha$ whenever the bound dominates the estimator's bias. For the ATT this gives $h_{\mathcal{G}} = h(B_{\mathcal{G}}, s_{\mathcal{G}})$, and under ex-ante S the same applies to the LATT, so $h_S = h(B_S, s_S)$. Both half-widths then share $\varphi(B, s) = s\,\text{cv}_\alpha(B/s)$, so $h_S < h_{\mathcal{G}}$ is the displayed inequality. Under data-driven selection $\hat{\theta}_S \mid \{\hat{S} = S\}$ is truncated Gaussian, so the honest LATT interval is the carved interval of Theorem 1 rather than the fixed-length (5). Its length is data-dependent and generally asymmetric, and its conditional expected length is infinite by Proposition 5, so no expected-length ordering is well-posed; the displayed threshold is therefore stated for the fixed-length intervals of ex-ante selection only. Fix $(s_{\mathcal{G}}, s_S, B_S)$ and set $c_0 = s_S\,\text{cv}_\alpha(B_S/s_S)$. The map $g(B) = s_{\mathcal{G}}\,\text{cv}_\alpha(B/s_{\mathcal{G}})$ is continuous and strictly increasing, from $g(0) = s_{\mathcal{G}}\,z_{1-\alpha/2}$ to ∞. Since $\text{cv}_\alpha : [0, \infty) \to [z_{1-\alpha/2}, \infty)$ is a strictly increasing bijection and when $s_S \geq s_{\mathcal{G}}$ we have $c_0/s_{\mathcal{G}} \geq (s_S/s_{\mathcal{G}})\,z_{1-\alpha/2} \geq z_{1-\alpha/2}$, so the equation $g(B^*_{\mathcal{G}}) = c_0$ has the unique solution displayed and $h_S < h_{\mathcal{G}} \iff B_{\mathcal{G}} > B^*_{\mathcal{G}}$. When $s_S < s_{\mathcal{G}}$ the argument $c_0/s_{\mathcal{G}}$ can fall below $z_{1-\alpha/2}$, $g(B^*_{\mathcal{G}}) = c_0$ has no nonnegative solution, and $h_S < h_{\mathcal{G}}$ already at $B_{\mathcal{G}} = 0$. Under precision weights, reweighting onto the retained set cannot lower the aggregate variance below the full set's optimum, so $s_S \geq s_{\mathcal{G}}$ and $B^*_{\mathcal{G}} > 0$ whenever $s_S > s_{\mathcal{G}}$. Under other weightings the ordering must be checked. $\square$